

Unequal Opportunities: Worker Adjustment After a Sectoral Collapse

Henry Redondo*

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Abstract

Workers displaced by the same sectoral shock in the same local labor market recover at different rates. I show that recovery varies systematically with how closely a worker's pre-shock skills match the composition of surviving local labor demand, a property I call skill–demand alignment. I construct a worker-level reallocation index from administrative records and study Spain's 2007–2012 construction collapse. Moving from the bottom to the top of the index reduces cumulative employment losses by 34 percent, yet leaves reemployment wages unchanged; additional evidence weighs against match-quality and ability-sorting explanations. The pattern is consistent with alignment operating through the arrival rate of acceptable offers. The employment gradient is primarily within-province, while earnings losses partly reflect the local wage structure of surviving demand.

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*Universidad de Alicante. E-mail: henry.redondo@ua.es. I am deeply indebted to Jan Stuhler for his guidance and advice throughout this project. I also thank Jesús Fernández-Huertas, Clara Santamaría, Marco Manacorda, Juan Dolado, María Castellanos, Álvaro Delgado, Lidia Cruces, and Camila Steffens for valuable discussions. Seminar participants at Universidad Carlos III de Madrid, University College London, and Queen Mary University of London provided helpful feedback. A previous version circulated as *From Bricklayers to Waiters: Reallocation in a Deep Recession*. All errors are my own.

1 Introduction

Even among workers displaced by the same sectoral shock in the same local labor market, reemployment trajectories differ. One literature emphasizes portable worker characteristics, including general and industry-specific human capital and job tenure (Jacobson et al., 1993; Davis and von Wachter, 2011; Lachowska et al., 2020; Gulyas and Pytka, 2025; Autor et al., 2014). A second highlights local labor market conditions (Bartik, 1991; Kline and Moretti, 2014; Yagan, 2019; Notowidigdo, 2020). Both isolate worker and place components of adjustment, leaving their interaction underexplored. I show that this within-market heterogeneity matters for reemployment speed, and that it matters most when a sector collapses permanently and geographic mobility is limited.¹ Workers can neither return to their sector nor move, so reemployment depends on whether their skills overlap with those demanded by sectors still hiring locally.

I study this question in the context of the collapse of Spain’s construction sector. Construction’s employment share fell from 13 to under 6 percent between 2007 and 2012, and the level of construction employment was still 59 percent below its 2007 peak in 2019. I exploit cross-province variation in exposure to the shock, measured as the proportional change in construction employment between 2007 and 2012. Four features of the episode make it informative about reallocation. First, the decline reflected structural overbuilding, so reallocation rather than recall is the relevant adjustment margin. Second, the collapse was triggered by an international sudden stop (Mendoza, 2010), so cross-province differences in exposure were largely determined by pre-recession sectoral specialization instead of contemporaneous worker decisions. Third, construction employed workers with a broad range of skill profiles, so otherwise similar workers differed in how well their skills matched the sectors still hiring locally. Fourth, the geographic mobility response was statistically indistinguishable from zero; workers’ adjustment therefore occurred within local labor markets, not across them. These features generate within-market variation in the quality of worker–market matches that aggregate local conditions do not capture.

To measure this match empirically, I construct a worker-level reallocation index, $RI^w(X_i, r)$, from administrative records, defined for a worker i with pre-shock characteristics X_i in province r . The index is a demand-weighted sum across non-origin sectors. Each sector’s contribution depends on three predetermined components: skill transferability from the origin sector, the sector’s local employment share, and pay relative to the origin sector. The weighting is motivated by a standard search-and-matching framework (Mortensen and Pis-

¹The absence of out-migration as a response to large labor market shocks is documented in other contexts, with adjustment occurring primarily within region (Autor et al., 2013; Dix-Carneiro and Kovak, 2017). These findings for Spain are consistent with this pattern.

sarides, 1994; Petrongolo and Pissarides, 2001). Because the index is built from the demand structure that survives locally after the shock, it captures the rate at which acceptable offers arrive and not the productivity of the resulting match. That distinction yields a testable implication, which I formalize in Appendix A. Productivity gains from reallocation should show up, at least in part, in wages, whereas greater exposure to employers still hiring need not raise wages. The index predicts reemployment but not wages conditional on reemployment, which is what one would expect if it shifts the arrival rate of offers, and it bounds the contribution of match productivity to the employment effect.

The empirical specification combines province-level exposure with the worker-level index, so identification comes from within-province differences in skill–demand alignment among observationally similar workers. This raises an obvious concern, namely that the index may simply proxy for unobserved ability. The wage results speak directly to it. If higher-index workers were more able, they should earn more once reemployed, but the index has no predictive power for wages conditional on reemployment (Section 5). Three further exercises offer additional evidence. The estimates are stable when I add province fixed effects and worker controls interacted with the regional shock; they are similar when I allow the relationship with the index to be non-linear; and pre-recession employment trends are parallel across the distribution of index values.

The shock–index interaction is 0.332 (s.e. 0.080) for cumulative employment and 0.002 (s.e. 0.003) for reemployment wages. Holding exposure fixed, moving from the bottom to the top of the index offsets 34 percent of cumulative employment losses, or about five months of additional employment over 2007–2012. The attenuation of the shock varies within-province. Including province fixed effects leaves the estimate essentially unchanged, from 0.332 to 0.322, so the gradient reflects mainly variation across workers within local labor markets. Cumulative earnings losses shows an equivalent gradient. That estimate is less precise, and it is sensitive to province fixed effects, indicating that it depends on the sectoral wage structure of local demand, which varies mainly across provinces.

Comparing the baseline index with two variants clarifies what it captures. A wage-excluded variant predicts employment losses nearly as precisely as the baseline, whereas a variant weighting demand by sectoral wage premia enters with the opposite sign. The contrast is informative. What matters is the alignment between a worker’s skills and the local sectoral mix, not the presence of high-paying jobs or sectoral diversity. The index therefore differs from existing measures of skill fit, such as skill remoteness (Macaluso, 2025) and the outside-options index of Caldwell and Danieli (2024). These measures treat transferability as a worker attribute defined against the pre-shock distribution of local jobs, and are well suited to studying match quality and outside options in equilibrium. It is also distinct

from geographic mismatch, which [Marinescu and Rathelot \(2018\)](#) find contributes little to aggregate unemployment in steady state. The relevant mismatch here is between a worker’s skills and the local demand that survives the shock.

The additive view treats the cost of displacement as separable into what the worker brings and what the local market offers. On that view, giving a low-skill worker portable human capital, or moving her to a stronger local labor market, should close most of the gap. It fails here, in the setting most favorable to it: one large shock, one sector, similar workers in the same local labor market. The worker–market pairing carries explanatory power beyond worker and place characteristics taken separately.

The result carries two implications. First, because alignment is measured before the shock, the index identifies *ex ante* which workers will reallocate slowly. Second, retraining toward sectors with high national demand pays only where local demand for those skills exists, so it is weakest in the markets where displacement bites hardest ([Manning and Petrongolo, 2017](#)). This interpretation may extend to other sectoral collapses, where place-level measures of labor-market slack can understate worker-level consequences when displaced workers exit into inactivity or disability instead of reallocating within a diversified local labor market ([Autor et al., 2013](#); [Beatty et al., 2007](#)).² Earnings, which additionally reflect the province-level wage structure of surviving demand, suggest a role for place-based policy alongside. A longstanding question is whether to target disadvantaged workers or disadvantaged places ([Austin et al., 2018](#)); here the two interact, and the match between them is what determines the cost of displacement.

I complement [Beaudry et al. \(2012\)](#), who show that local industrial composition shapes wages in the steady state. Applied to post-displacement dynamics, the same logic governs how fast workers reemploy, not what they earn once they do. The result also speaks to the sorting literature ([Lise and Postel-Vinay, 2020](#)), where worker–market complementarities are usually studied through who matches with whom in equilibrium; here they determine how quickly displaced workers find any acceptable match after a shock disrupts that equilibrium. Finally, it refines the Jacobs view that industrial variety *per se* confers resilience ([Glaeser et al., 1992](#)). The relevant margin is not diversity but alignment, consistent with the distinction between related and unrelated variety in evolutionary economic geography ([Frenken et al., 2007](#)), identified here at the level of the individual involuntary separator and not as a regional growth correlation.

²Examples include the decline of UK coal mining after the 1984–85 pit closures ([Beatty and Fothergill, 1996](#); [Beatty et al., 2007](#)) and the contraction of US manufacturing following Chinese import competition after 2001 ([Autor et al., 2014](#)). Similar dynamics may arise from climate-transition policies in carbon-intensive industries and the adoption of AI and robotics in routine-task occupations, which likewise concentrate displacement by occupation and place.

The remainder of the paper proceeds as follows. Section 2 discusses the collapse of Spain’s construction sector. Section 3 describes the data, and Section 4 introduces the reallocation index. Section 5 documents the labor market consequences of the shock, while Section 6 shows how worker–demand alignment mitigates displacement costs and the underlying adjustment mechanisms, and Section 7 presents a counterfactual policy exercise. Section 8 reports robustness checks. Section 9 concludes.

2 Setting: Spain’s Construction Collapse

Between 2007 and 2012, Spain’s construction sector shed nearly 1.2 million workers, contracting from 13 percent of total employment to below 6 percent (Figure 1). The contraction was structural, not cyclical. During the boom Spain had built housing at rates far exceeding long-run demand, driven by interest rate convergence following EU accession and amplified by domestic credit expansion. Once credit conditions normalized, employment remained 59 percent below its 2007 peak as late as 2019. U.S. construction, by comparison, largely recovered by 2015 as an underlying housing shortage reasserted itself (Online Appendix Figure OA.1).

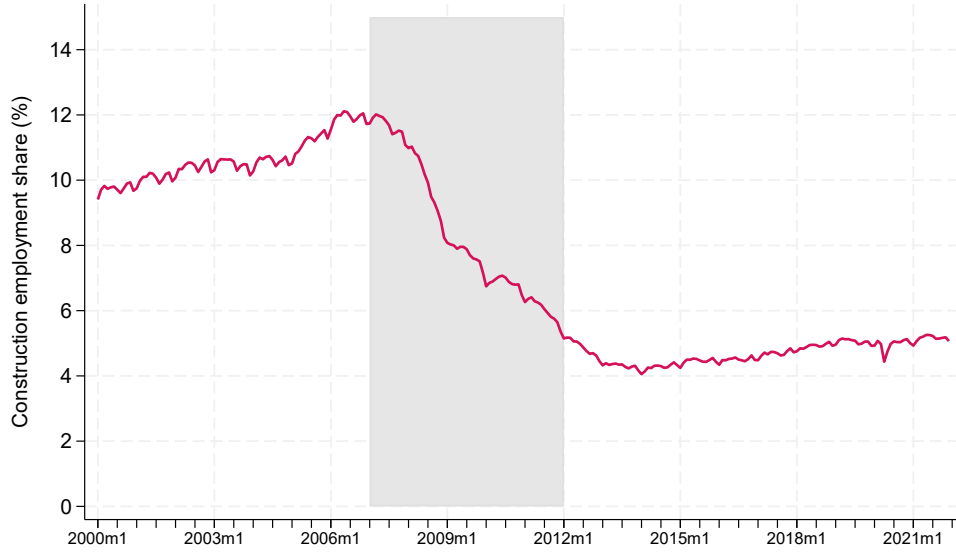
Three of these properties deserve elaboration, because they carry the identification. First, the shock originated in a sudden stop of international capital inflows (Mendoza, 2010; Ben-tolila et al., 2012), so cross-province differences in severity trace to pre-recession sectoral specialization, plausibly predetermined with respect to workers’ post-2007 trajectories; measuring exposure in levels rather than shares yields nearly identical results (Table 1, Column 4). Second, interregional mobility in Spain is low, so adjustment occurs predominantly within local labor markets and the shock is not confounded by selective sorting across provinces. Third, construction spans the full skill distribution within a single sector and province, so the same provincial shock generates heterogeneous reallocation prospects across workers, enabling within-province comparisons that hold local labor demand fixed. Sections 5.3 and 6.2 provide complementary evidence that, conditional on controls, the provincial variation in exposure is uncorrelated with pre-recession worker outcomes.

The contraction exhibited pronounced regional heterogeneity (Appendix Figure OA.8). Provincial construction employment losses ranged from around 20 percent in several northern provinces to more than 60 percent in the south, with an interquartile range of 10.3 percentage points in the provincial employment share decline of the construction sector.

Figure OA.9 tracks workers employed in construction in 2007 and illustrates the breadth of the displacement.³ By 2015, roughly 30 percent of former construction workers were

³The patterns in Figure OA.9 are not driven by compositional differences across skill groups or by pre-

Figure 1: Employment share of workers in the Spanish construction sector, 2000-2021



Notes: Employment share of Spain’s construction sector, January 2000–December 2021. Monthly observations are based on workers aged 20–60. The shaded area denotes the Great Recession period (2007–2012).

employed in other sectors, while the share who relocated across provinces remained below 8 percent.

3 Data

The primary data source is the *Muestra Continua de Vidas Laborales* (MCVL), a longitudinal administrative dataset containing a 4 percent random sample of Spain’s Social Security affiliates. The MCVL links Social Security records to income tax filings and the population register. Spell-level records include employment and unemployment episodes and report earnings, contract type, firm size, two-digit industry, occupation, part-time status, workplace location, and reason for job separation.⁴ Individual-level information includes gender, educational attainment, date and province of birth, citizenship, and household composition.

Earnings are available from two sources. Social Security records provide monthly earnings back to 1980 but are subject to annually adjusted contribution floors and ceilings; tax records, linked from 2006 onward, are uncensored. Censoring is minimal in the construction

existing sectoral mobility trends. Appendix Figure OA.10 replicates the analysis for high-skilled construction workers and finds qualitatively identical dynamics, though with smaller employment losses, indicating that the broad reallocation pattern is not specific to low-skill occupations. Appendix Figure OA.11 applies a placebo shock in 2003 and finds no differential pre-period trajectories, inconsistent with the interpretation that Figure OA.9 reflects pre-recession differences in sectoral composition.

⁴Contract information is recorded at the spell level, including exact start and end dates.

sample—only 0.6 percent of worker-month observations are top- or bottom-coded—so I use tax records as the primary measure from 2006 onward, but complement it with the Social Security earnings data.⁵ The uncensored tax records permit analysis across the full earnings distribution. All earnings are deflated to 2009 prices using the national CPI.

I construct a monthly panel spanning 2000–2019. The complete sample includes all workers aged 18–60 affiliated with Social Security across all 50 Spanish provinces, excluding Ceuta and Melilla, and is used for descriptive analysis.⁶ I apply four restrictions to the estimation sample. First, I require at least six months of construction employment during 2005–2006, selecting workers with established sectoral attachment (average construction tenure in 2007 was 11 months) and excluding marginal spells from the boom’s final expansion.⁷ Second, I require the worker to be observed employed in construction in at least one month of 2007, so that all sample members were active in the sector at the onset of the shock. Third, I restrict the sample to native workers. Foreign-born workers constituted 27 percent of construction employment in 2007 and exhibit higher rates of return migration during recessions, which the administrative records do not capture and which would generate differential attrition in high-exposure provinces. The estimates are nonetheless robust to controlling for baseline foreign-worker shares (Section 8). Fourth, I keep workers aged 20–50 in 2007, to isolate labor market adjustment from retirement transitions. Workers are followed from January 2007 through December 2019, with the main analysis ending in December 2012. The resulting estimation sample comprises 34,000 workers; summary statistics are reported in Appendix Table OA.1.

4 The Reallocation Index

Workers displaced by the same shock do not face the same prospects. A bricklayer and an electrician laid off in the same province face different sets of available jobs, because their skills transfer differently across sectors and local demand differs across those sectors. Standard regional exposure designs assign a single measure to everyone in a province and so treat these workers as equally exposed. The index developed here instead exploits variation within provinces. I explain the intuition behind its form and state the predictions tested

⁵Bonhomme and Hospido (2017) document that earnings differences between tax and Social Security records are concentrated above the 90th percentile. Workers in the Basque Country and Navarre are covered by Social Security records only due to separate fiscal arrangements. I compute days worked within each calendar month from the underlying employment spells and convert annual tax earnings into daily equivalents to ensure consistency across sources.

⁶Ceuta and Melilla are excluded due to their distinct labor market structures and fewer than 200 construction workers in 2007.

⁷Results are robust to alternative thresholds of 3, 9, and 12 months (Appendix Table C6).

below; Appendix A gives the formal framework.

4.1 A Simple Framework

Consider a worker searching across local sectors after losing a construction job. Whether she returns to work quickly depends on how often she receives an offer she is willing to accept. A standard directed search framework (Appendix A) implies that acceptable offers arrive at a higher rate when her skills transfer more readily to a destination sector, when that sector accounts for a larger share of local employment, and when it pays more relative to her origin sector.⁸ The index multiplies these three terms within each sector and sums across sectors.

Let X_i denote a vector of worker i 's observable characteristics—occupation, foreign-born status, education, age, and gender—and r her province of residence. The wage-weighted reallocation index is

$$RI^w(X_i, r) = \sum_{j \neq cs} \hat{\phi}_j(X_i) \cdot \frac{s_{jr}}{\bar{s}_j} \cdot \frac{w_j}{w_{cs}}, \quad (1)$$

where $\hat{\phi}_j(X_i)$ is the predicted probability that a worker with characteristics X_i is employed in sector j ; $s_{jr}/\bar{s}_j$ is the relative concentration of sector j in province r ; and w_j/w_{cs} is the ratio of sector j 's median wage to the construction-sector median wage, with all measured in 2006. The measure, $RI^w(X_i, r)$ (hereafter RI_i), summarizes the earnings-weighted alignment between a worker's skill profile and the structure of local labor demand.

Each component reflects one of the three determinants discussed above. The match probability $\hat{\phi}_j(X_i)$ stands for skill compatibility, the chance that a worker like i enters sector j conditional on leaving construction. The specialization ratio $s_{jr}/\bar{s}_j$ stands for local opportunity. It gives more weight to sectors that are overrepresented in the worker's province, while preventing large national sectors from dominating mechanically, and it excludes construction's own local share, so the index measures prospects *out of* the declining sector. The wage ratio w_j/w_{cs} stands for wage adequacy, since offers paying far below pre-displacement earnings are less likely to be accepted and so contribute less.⁹

All three inputs are measured before the recession, ensuring each component is predetermined with respect to post-2007 outcomes. Match probabilities are estimated from a multinomial logit of sectoral employment on hiring spells of non-construction workers between 2002 and 2006, so the probabilities reflect sector entry opportunities and not incumbency

⁸Formally, labor-market tightness is assumed to increase with local sectoral specialization and the acceptance probability to increase with the relative wage. These restrictions are detailed in Appendix A.

⁹The wage ratio proxies for the probability that an offer is accepted, not for match surplus. A uniform-weight variant that replaces w_j/w_{cs} with unit weights therefore ignores variation in the attractiveness of potential offers. Section 6.3 finds that the predictive content of the index derives primarily from skill transferability and not from differential access to high-wage sectors.

patterns. Local sectoral shares s_{jr} are measured at the province level in 2006, excluding construction; sector median wages w_j are computed nationally from 2006 earnings.¹⁰

Two concerns remain. The first is whether the skill–sector mapping continued to operate during the downturn. Because $\hat{\phi}_j(X_i)$ reflects relatively stable structural features of how worker characteristics match sector requirements, it should be largely unaffected by the contraction in employment levels. Consistent with this, re-estimating the logit on post-recession hiring spells (2010–2015) yields an index highly rank-correlated with the baseline ($\rho = 0.873$). The second is sorting. If high-ability workers sort into provinces with favorable sectoral composition, the index might absorb unobserved quality instead of local opportunity. Conditioning on observable characteristics absorbs the part of sorting that observables can explain, and the abruptness of the capital-flow reversal makes anticipatory sorting implausible (Section 2). What remains is unobserved ability. If high-index workers were more able, their employment paths should already have been diverging before the recession. They were not (see Appendix Figure B1).

The model delivers asymmetric predictions across the two margins of adjustment. The offer arrival rate depends on within-province sectoral shares, so its cross-worker variation is identified independently of province-level wages; the expected reemployment wage, by contrast, depends on the absolute wage level of destination sectors in province r . Worker-level alignment should therefore predict reemployment speed within a province after absorbing all province-level variation, while the earnings gradient partially reflects between-province differences in the wage level of surviving sectors. Section 6.3 tests both predictions.

4.2 Properties: Within- versus Between-Province Variation

The index varies along two dimensions. Across provinces, the same worker profile maps into different opportunity sets as local sectoral composition changes; within a province, different worker profiles access different subsets of the same local labor market. The within-province dimension dominates. The within-province standard deviation of the index is 2.3 times the between-province standard deviation in the estimation sample.

The baseline index isolates the channel that dominates in the Spanish setting. When workers cannot relocate and the origin sector has ceased hiring, skill–demand alignment within the local labor market is the primary margin of adjustment. Two extensions in Appendix A.4 relax these restrictions. When geographic mobility is feasible, access to neighboring labor markets partially substitutes for local alignment. When the contracting sector retains residual demand, the index converges toward a conventional outside-options mea-

¹⁰National rather than province-specific wages capture sector-specific skill requirements common across regions and avoid local cost-of-living differences unrelated to job quality.

sure. Both extensions clarify the scope conditions under which the baseline index captures the relevant opportunity set.

Workers with greater reallocation opportunities should return to work more quickly, experience smaller earnings losses, and sort into sectors compatible with their skills and well-represented locally. A further implication is that aggregate measures of sectoral diversity should contain little additional information once worker-specific opportunities are accounted for; Section 6.3 assesses this prediction directly.

5 The Cost of the Construction Collapse

5.1 Empirical Strategy and Identification

Identification exploits cross-province variation in exposure to the 2007–2012 construction contraction. I compare observably similar workers in provinces that experienced differential declines in construction activity, using the decline in provincial construction employment share as the shock measure. The employment share decline is interpretable as a labor demand shock. Regional employment fluctuations are predominantly demand-driven in the medium run (Blanchard and Katz, 1992), and the trigger here, an abrupt tightening of international credit conditions beginning in 2007–2008, was common across provinces (Section 2). Province-level variation in severity therefore reflects pre-recession differences in sectoral composition, not worker behavior or local labor supply.

Because adjustment occurred over several years, I focus on cumulative earnings over 2007–2012, normalized by average annual earnings in 2005–2006 (Autor et al., 2014; Yagan, 2019). This measure captures the combined effects of wage changes and non-employment over the full adjustment period. The baseline specification is:

$$y_i = \theta \text{Shock}_{r(i)} + X_i' \beta + \varepsilon_i, \quad (2)$$

where $\text{Shock}_{r(i)}$ denotes the proportional decline in construction’s employment share between 2007 and 2012 in worker i ’s pre-recession province of residence, r .¹¹ The vector X_i contains predetermined worker characteristics measured in 2007, including age interacted with gender and education, occupational skill group, tenure, labor-market experience and its square, indicators for fixed-term and part-time contracts, and baseline earnings quintiles. It also includes province-level controls measured in 2006: the unemployment rate, construction’s initial employment share, a Bartik demand shifter for non-construction sectors, and a

¹¹Let s_{rt} denote construction’s share of total employment in province r and year t . The shock is defined as $-(s_{r,2012} - s_{r,2007})/s_{r,2007}$, such that larger values indicate a more severe construction collapse.

Herfindahl-Hirschman index of non-construction sectoral concentration.

I also estimate an IV specification. I treat these results as corroborative rather than primary, since instrumenting the shock reduces the precision of the heterogeneity estimates that are the paper’s focus. I instrument the provincial shock with average annual growth in construction employment during 2000–2006: provinces with faster pre-recession expansion accumulated greater sectoral exposure, generating differential vulnerability to the subsequent collapse. The instrument satisfies the relevance condition: a one-percentage-point increase in pre-recession construction growth predicts a 1.287-percentage-point larger decline in the construction employment share between 2007 and 2012 (Appendix Figure OA.5), with a Sanderson–Windmeijer F -statistic of 12.44. Weak-instrument-robust Anderson–Rubin inference rejects the null of no effect, with a 95 percent confidence interval of $[-6.08, -1.27]$ ($p = 0.017$).¹²

The main threat to the exclusion restriction is that pre-recession construction growth proxies for broader provincial characteristics correlated with post-recession outcomes through channels other than the shock itself. Two pieces of evidence support conditional independence. First, the first-stage coefficient is virtually unchanged when conditioning on pre-recession growth in mortgage issuance, mortgage values, and immigrant population (Table C2), indicating the instrument does not operate through differential credit expansion, housing-market dynamics, or labor-supply composition. Second, reduced-form regressions of cumulative earnings, annual earnings, and cumulative days worked during 2003–2006 on the instrument yield small, statistically insignificant coefficients across all three outcomes (Table C1), providing no evidence of differential pre-trends in labor market trajectories.

5.2 Earnings and Employment Losses

Baseline results. Table 1 reports estimates of equation (2). Controlling only for life-cycle earnings patterns (Column 1), a one-standard-deviation increase in the provincial shock (8.9 percentage points) reduces cumulative earnings by roughly 4.5 months of pre-recession pay over 2007–2012.¹³ Adding job characteristics—occupational skill group, tenure, experience, and contract type—reduces the coefficient to -2.793 (Column 2). The 34 percent attenuation indicates that more-exposed provinces held workers with less transferable job characteristics, so part of the raw gap reflects worker composition.

Column 3 adds province-level labor-market controls and is the preferred specification. Moving from the 25th to the 75th percentile of the exposure distribution reduces cumulative

¹²Wild-bootstrap inference yields similar conclusions (see Section 8), addressing concerns associated with inference based on 50 province clusters (Cameron et al., 2008).

¹³ $4.241 \times 0.089 \approx 0.377$ times pre-recession annual earnings, equivalent to approximately 4.5 months.

Table 1: Cumulative Earnings Impact of the Construction Employment Decline, 2007–2012

	OLS				IV
	(1)	(2)	(3)	(4)	(5)
<i>Dependent variable: Normalized cumulative earnings, 2007–2012</i>					
Shock	−4.241*** (0.464)	−2.793*** (0.327)	−2.067*** (0.292)	−2.026*** (0.291)	−3.111*** (0.781)
Observations	34,000	34,000	34,000	34,000	34,000
Age × Gender × Education	✓	✓	✓	✓	✓
Job characteristics		✓	✓	✓	✓
Regional controls			✓	✓	✓

Notes: The shock is defined as the proportional change in the construction sector’s employment share at the province level between 2007 and 2012. Column (4) uses the proportional change in the number of construction workers by province as an alternative measure. Column (5) reports instrumental variables estimates, instrumenting the shock with average annual growth in construction employment over 2000–2006. Job-level controls include occupational skill group, indicators for part-time and fixed-term contracts, earnings quintiles, and tenure, and experience fixed effects. Regional controls include the local unemployment rate, the initial construction employment share, a Bartik measure of demand shifts in non-construction sectors, and the Herfindahl–Hirschman Index (HHI) for non-construction sectors. Standard errors are clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

earnings by approximately 21 percent of pre-recession annual earnings.¹⁴ Column 4 shows nearly identical results when exposure is measured in levels rather than shares, confirming the findings are not an artifact of changes in aggregate provincial employment.

The IV estimate of −3.111 (Column 5) is somewhat larger in absolute value than the corresponding OLS estimate, but the two remain economically similar. This pattern is consistent with attenuation bias in the OLS specification arising from measurement error in the exposure measure. It may also reflect a local average treatment effect if the instrument places greater weight on provinces where pre-recession construction growth translated more strongly into subsequent labor market exposure. The IV estimate is larger than its OLS counterpart but retains the same sign and significance, a pattern consistent with attenuation in the OLS specification rather than with confounding that would overstate the effect.

To understand the sources of these losses, I first decompose aggregate earnings effects into their extensive and intensive margins.

Intensive and extensive margins. Table 2 decomposes the cumulative earnings effect into an extensive margin, days employed over 2007–2012 expressed in years (Column 2), and an intensive margin, average monthly earnings conditional on employment, normalized by the 2005–2006 baseline used for cumulative earnings (Column 3).

¹⁴The interquartile difference in employment share declines is 0.103 (75th percentile: 0.620; 25th: 0.517). Evaluated at the estimated coefficient: $-2.067 \times 0.103 \approx 0.213$ times annual pre-recession earnings.

Reduced employment duration accounts for roughly 73 percent of the total earnings loss,¹⁵ making it the dominant margin. Workers in more-exposed provinces accumulate fewer days of work, not lower wages when employed. The conditional-wage effect is small, consistent with [Lachowska et al. \(2020\)](#), who find that nonemployment duration accounts for the bulk of long-term earnings losses among displaced workers in the US.

The remaining 27 percent is consistent with modest wage losses upon reemployment. Because construction paid relatively high pre-recession wages and even induced an increased dropout rate ([Lacuesta et al., 2020](#)), workers reemployed in other sectors likely earned less even in comparable positions. These losses appear to reflect moves into sectors paying somewhat below construction, and not substantial downward mobility in the occupation-wage distribution.

Because earnings losses are driven largely by non-employment, adjustment depends on the availability of suitable reemployment opportunities. Workers whose skills align more closely with the sectors present in their local labor market should receive acceptable offers more quickly, spend less time out of work, and experience smaller cumulative losses. The reallocation index developed in Section 4 is designed to capture this variation in reemployment opportunities.

Table 2: Decomposition of Earnings Losses: Employment versus Wages

	(1) Cumulative Earnings	(2) Cumulative Days Employed	(3) Avg. Earnings
Shock	-2.067*** (0.292)	-1.627*** (0.159)	-0.014* (0.008)
Observations	34,000	34,000	34,000
Controls	✓	✓	✓

Notes: All specifications include the preferred set of individual and regional controls from Column (3) of Table 1. Robust standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

¹⁵Cumulative normalized earnings can be expressed as $Y_i = E_i \times 12 w_i$, where E_i denotes years of employment and w_i is average monthly earnings divided by average annual pre-displacement earnings. A first-order decomposition of the shock effect yields $\partial Y / \partial S \approx 12 \bar{w} \cdot (\partial E / \partial S) + 12 \bar{E} \cdot (\partial w / \partial S)$, where $\bar{E} = 3.914$ and $\bar{w} = 0.093$ are sample means. Evaluating: the employment channel contributes $12 \times 0.093 \times (-1.627) = -1.816$ and the wage channel contributes $12 \times 3.914 \times (-0.014) = -0.658$, implying shares of 73 and 27 percent, respectively. The linear approximation overshoots the estimated total (-2.067) because it omits the covariance between E_i and w_i .

5.3 Dynamic Adjustment and Persistence

The cumulative estimates leave the timing of adjustment open. Year-by-year effects of the construction shock trace the trajectory of earnings and employment losses:

$$y_{it} = \text{Shock}_{r(i)} \cdot \beta_t + X'_i \Delta_t + \varepsilon_{it}. \quad (3)$$

The dependent variables are annual earnings normalized by 2007 earnings and a binary employment indicator, and the pre-recession coefficients provide a direct test of parallel trends.

Figure 2 plots the estimated $\hat{\beta}_t$. Pre-recession estimates are small and statistically indistinguishable from zero for both outcomes, supporting the identifying assumption. After 2007, employment probabilities in more-exposed provinces decline sharply and remain depressed for several years, while conditional wages stay near pre-recession levels throughout, reinforcing the decomposition’s finding that non-employment, not wage compression, drives cumulative losses.

At the 2012 trough, a one-standard-deviation larger shock lowers the annual employment probability by roughly 1.5 percentage points.¹⁶ The gap narrows rapidly. The coefficient falls to -0.035 by 2014, a 77 percent compression, and reaches $+0.007$ by 2015, statistically indistinguishable from zero, so workers across the exposure distribution converge to similar annual employment rates within eight years.

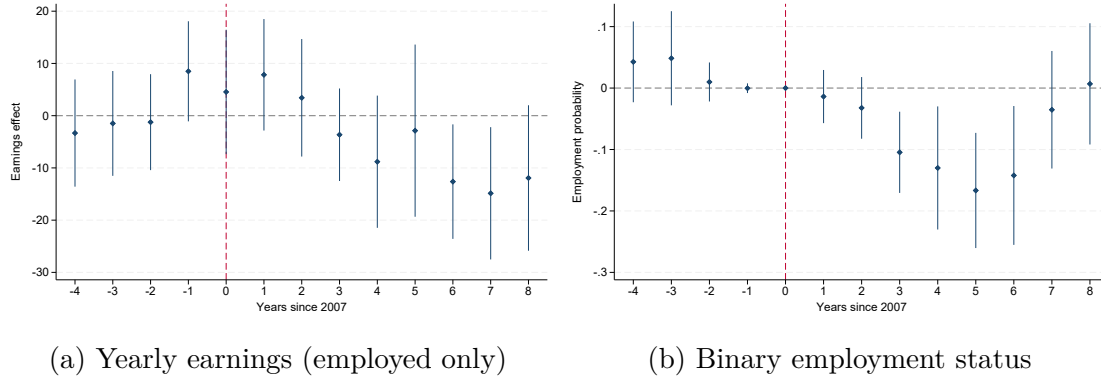
Convergence in annual employment rates coexists with substantial differences in cumulative histories. Workers in more exposed provinces spent considerably more time out of work, generating losses that subsequent recovery cannot fully reverse. Since construction itself did not rebound, this convergence reflects reallocation into other sectors rather than re-entry into the original industry. Section 6 documents this adjustment process and shows that migration plays little role.

Combined with the absence of meaningful wage effects, these results indicate that long-run recovery depends primarily on how quickly workers reenter employment. The next results examine which worker characteristics are associated with faster adjustment and use this evidence to motivate a measure of reallocation capacity.

Heterogeneity. The average effects conceal substantial heterogeneity. To characterize this heterogeneity, I interact the provincial shock with baseline worker characteristics (Tables OA.7 and OA.8). Workers with less labor-market experience, lower pre-recession earnings, and younger age experience significantly larger cumulative earnings losses follow-

¹⁶ $0.167 \times 0.089 \approx 0.0149$. At the interquartile difference (10.3 percentage points), the implied employment probability gap is $0.167 \times 0.103 \approx 0.017$ in 2012.

Figure 2: Year-by-Year Impact of the Construction Employment Decline



Notes: Coefficients from equation (3). Additional controls as in Column (3) of Table 1. Confidence intervals based on robust standard errors clustered at the province level.

ing the collapse. Estimated effects are also larger for workers with lower education and shorter tenure, although these differences are less precisely estimated. Occupational skill group is the main exception. Conditional on other characteristics, low-skilled workers do not experience significantly larger losses than their high-skilled counterparts.

Figure OA.14 documents the magnitude of these differences. Workers younger than 30 at the recession's onset experienced the largest losses of any age group, consistent with their disproportionate concentration in temporary positions with limited transferability outside construction. Less-experienced workers and those in the lower half of the pre-recession earnings distribution likewise suffered larger losses. The skill-group null sharpens the picture. Once earnings and experience are controlled for, occupational classification adds little, suggesting that accumulated human capital, not occupational category, is the more relevant dimension for reallocation prospects.

These patterns motivate but do not identify the mechanism. Observable characteristics leave substantial residual heterogeneity unexplained. Within any age, education, or earnings cell, workers in the same province face similar shock exposure yet experience markedly different adjustment trajectories. Section 6 uses the reallocation index to explain this within-cell variation.

6 Skill–Demand Alignment and Adjustment Heterogeneity

Because earnings losses run primarily through prolonged non-employment rather than wage penalties conditional on reemployment (Section 5), the speed of reentry accounts for the bulk

of post-displacement costs. I now ask whether that speed depends on the alignment between worker skills and locally available non-construction sectors. If so, workers facing less overlap with local demand should sustain larger cumulative losses regardless of aggregate provincial conditions.

6.1 Reallocation Capacity and Earnings Losses

I augment equation (2) with the reallocation index:

$$y_i = \theta \text{Shock}_{r(i)} + \gamma \text{RI}_i + \delta \text{Shock}_{r(i)} \times \text{RI}_i + X_i' \beta + \varepsilon_i \quad (4)$$

where RI_i is the reallocation index constructed in Section 4, which varies across workers within the same province, and $\text{Shock}_{r(i)}$ as in equation (2).

Table 3: Labor Market Impact of the Construction Employment Decline, 2007–2012

	Employment		Cumulative Earnings		Avg. Earnings	
	(1)	(2)	(3)	(4)	(5)	(6)
Shock	−1.617*** (0.156)	−1.616*** (0.156)	−2.045*** (0.286)	−2.045*** (0.278)	−0.013* (0.007)	−0.013* (0.007)
Reall. Index	0.045** (0.020)	−0.149*** (0.051)	0.096*** (0.028)	−0.105 (0.098)	0.001** (0.000)	0.000 (0.002)
Shock × Reall. Index		0.332*** (0.080)		0.344** (0.162)		0.002 (0.003)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: Dependent variables are cumulative days employed annualized (Columns 1–2), normalized cumulative earnings (Columns 3–4), and average monthly earnings conditional on employment (Columns 5–6), each measured over 2007–2012 and normalized by the worker’s average annual earnings in 2005–2006. The shock measures the proportional change in the construction sector’s employment share at the province level between 2007 and 2012. The reallocation index, constructed using pre-recession data, measures the alignment between each worker’s skill profile and the sectoral composition of local labor demand outside construction. Odd-numbered columns include the reallocation index as a main effect; even-numbered columns add its interaction with the shock. All specifications include worker, job, and regional controls from Column 3 of Table 1. Standard errors are clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Consistent with the predictions of Section 4, the reallocation index is positively associated with cumulative days employed (Column 1 of Table 3), implying that workers with stronger reallocation prospects return to work more quickly. The key coefficient is the interaction between the index and the shock, which captures whether worker-specific reallocation opportunities mitigate the effects of a given provincial contraction. At the 75th percentile

of the provincial shock distribution, a one-standard-deviation improvement in the index attenuates the cumulative employment loss by 20.6 percent of the shock-induced employment penalty.¹⁷

The interaction is also positive for cumulative earnings (0.344, Column 4), offsetting approximately 16.8 percent of the earnings loss at a given level of exposure; this earnings gradient is less robust than the employment one and, as shown below, operates largely between rather than within provinces.¹⁸

The employment interaction equals 0.332 (0.080), whereas the conditional-wage interaction is 0.002 (0.003), statistically and economically negligible. Since 27% of cumulative earnings losses at the mean run through the conditional-wage margin, a match-quality mechanism operating in proportion to the shock’s margin structure would predict a wage interaction of roughly 0.090—more than four times the estimated effect.¹⁹

Because the conditional-wage sample is restricted to the reemployed—and the index predicts who returns to work—the null could in principle reflect selection rather than a true absence of wage effects. Since every worker is observed employed at least once during this period, the sample is not subject to the never-employed selection margin. To probe this, I re-estimate the wage interaction while progressively raising the minimum employment-spell threshold that defines reemployment, from the 25th to the 75th percentile of the cumulative-days distribution (Table OA.5). Tightening the threshold shifts the composition of the reemployed sample toward more strongly attached workers; if selection were generating the null, the wage interaction would move with it. It does not—the interaction remains statistically and economically negligible across all thresholds.

This divergence is difficult to reconcile with a match-quality account, in which high-index workers access intrinsically better-paying sectors and wage gains accompany employment gains. Instead, the index mitigates earnings losses by accelerating reemployment, not by raising wages conditional on reemployment, capturing access to suitable employment opportunities and not the quality of the eventual match. This is consistent with evidence that displaced workers, particularly during large recessions, accept reemployment in lower-paying occupations rather than waiting for quality matches (Huckfeldt, 2022), reinforcing that the binding constraint is access to any suitable job, not the wage premium of the eventual one.

To assess whether the moderating role of reallocation capacity varies systematically across

¹⁷The employment penalty at the 75th percentile of the shock is $-1.616 \times 0.620 \approx -1.002$; the attenuation from a one-standard-deviation improvement in the index is $0.332 \times 0.620 \approx 0.206$.

¹⁸The earnings penalty at the 75th percentile of the shock is $-2.045 \times 0.620 \approx -1.268$; the attenuation is $0.344 \times 0.620 \approx 0.213$, yielding $0.213/1.268 \approx 16.8$ percent.

¹⁹I establish this formally using a two one-sided tests (TOST) equivalence procedure. The 90% confidence interval on the wage interaction, $[-0.003, 0.007]$, falls well within the equivalence bounds $[-0.090, 0.090]$, ruling out effects of that magnitude at the 5% level.

Table 4: Sectoral Composition and the Consequences of the Construction Collapse: Heterogeneity by Reallocation Index Quartile

	Employment		Cumulative Earnings		Avg. Earnings	
	(1)	(2)	(3)	(4)	(5)	(6)
Shock	−1.627*** (0.159)		−2.067*** (0.292)		−0.014* (0.008)	
<i>Shock × Reallocation Index Quartile:</i>						
Q1 (lowest alignment)		−1.932*** (0.237)		−2.462*** (0.369)		−0.021** (0.010)
Q2		−1.710*** (0.215)		−2.039*** (0.368)		−0.000 (0.014)
Q3		−1.504*** (0.257)		−2.022*** (0.381)		−0.016* (0.009)
Q4 (highest alignment)		−1.268*** (0.235)		−1.590*** (0.507)		−0.011 (0.013)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: Dependent variables are cumulative days employed annualized (Columns 1–2), normalized cumulative earnings (Columns 3–4), and average monthly earnings conditional on employment (Columns 5–6), each measured over 2007–2012 expressed relative to the worker’s baseline average annual earnings in 2005–2006. Columns (1), (3), and (5) reproduce the baseline shock coefficient from Table 1. Columns (2), (4), and (6) introduce interactions between the shock and quartile indicators for the reallocation index, allowing the shock coefficient to vary across the index distribution. All specifications include worker, job, and regional controls from Column (3) of Table 1. Standard errors are clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the index distribution, I interact the shock with quartile indicators:

$$y_i = \sum_{k=1}^4 \beta_k Q_i^k \cdot \text{Shock}_{r(i)} + X_i' \Delta + \varepsilon_i, \quad (5)$$

where Q_i^k indicates that worker i belongs to quartile k of the index distribution. If reallocation capacity governs adjustment, shock coefficients should decline across quartiles, with better-aligned workers losing less under the same shock.

The employment gradient is monotonic and precisely estimated (Table 4). Workers in the highest quartile of the reallocation index are less affected by the construction shock than those in the lowest quartile. The estimated employment penalty declines by 34 percent across quartiles, from -1.932 in Q1 to -1.268 in Q4. An F -test rejects equality of the Q1 and Q4 coefficients ($p < 0.05$). The gradient is stronger at the tails of the distribution. The Q1–Q2 difference (0.222) and the Q3–Q4 difference (0.236) both exceed the Q2–Q3 difference (0.206). At the 75th percentile of the provincial shock distribution, these estimates imply a

loss of 1.198 years of employment for a worker in the bottom quartile of the index, compared with 0.786 years for an otherwise identical worker in the top quartile—a difference of 0.412 years, or nearly five months.²⁰

The cumulative earnings gradient runs in the same direction but, as the model predicts, operates largely between provinces rather than within them. The earnings penalty falls by 35 percent across quartiles, from -2.462 in Q1 to -1.590 in Q4. By contrast, conditional wages exhibit no monotonic relationship with the reallocation index. This pattern accords with the interpretation developed in Section 4. The index captures access to alternative employment opportunities through worker–demand alignment, not wage-setting conditional on reemployment. Workers in better-matched local labor markets return to work more quickly and spend more time employed, but do not systematically obtain higher wages once reemployed. Differences in search frictions, monopsony power, or sector-specific wage premia may influence accepted wages without materially affecting the speed or likelihood of reallocation.

This quartile spread exceeds the variation associated with any single observable in Section 5.3, including the education and age-below-30 contrasts.²¹

The same ordering appears under the wild bootstrap. The employment interaction is highly significant ($p = 0.000$), the earnings interaction weaker ($p = 0.071$), and the wage interaction insignificant throughout (Section 8). The index thus captures variation in the likelihood that displaced workers are reemployed in surviving sectors, rather than systematic differences in match quality, access to higher-paying industries, or sector-specific wage premia.

Two workers identical in age, education, and pre-recession earnings still face different prospects when local sectoral structure suits one better, and the index isolates this residual component that broad demographic groupings do not capture. Whether this within-province, within-shock heterogeneity translates into faster sectoral reentry is examined in Section 6.2.

6.2 Sectoral Reallocation Under Geographic Immobility

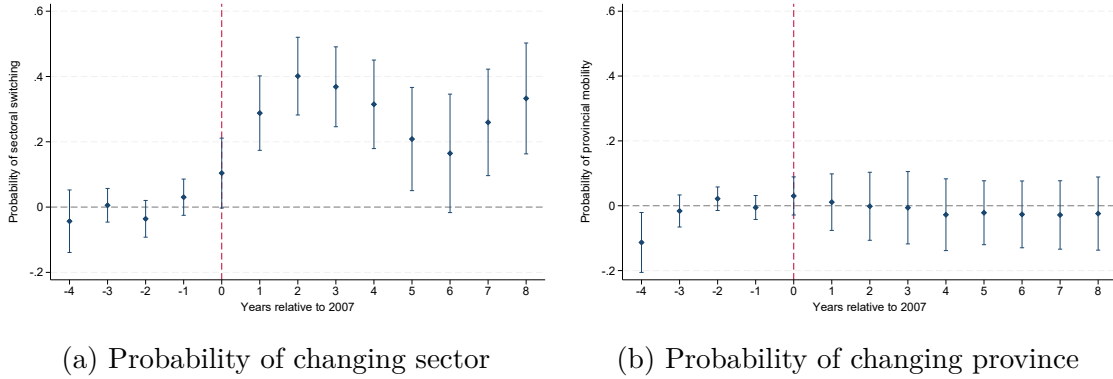
Displaced workers can adjust along two margins: moving to another province or leaving construction for another sector. Replacing cumulative earnings in equation (2) with a linear probability model for each, Figure 3 plots year-by-year shock coefficients over 2003–2015.²²

²⁰Penalties at the 75th percentile of the shock are $-1.932 \times 0.620 \approx -1.198$ years (Q1) and $-1.268 \times 0.620 \approx -0.786$ years (Q4).

²¹The comparison is approximate. The subgroup regressions in Table OA.8 use split samples, whereas the quartile interactions use the full sample.

²²Figure OA.16 presents the same analysis without conditioning on pre-recession average mobility. The results are qualitatively unchanged, although the estimates are somewhat less precise.

Figure 3: Probability of Sectoral versus Geographic Reallocation



Notes: Panel (a) reports the probability of transitioning out of the construction sector into a different sector; Panel (b) reports the probability of residing in a province different from the 2007 province of residence. The sample and control variables follow the baseline specification in Table 1. Both panels include the pre-period (2004–2006) average of the relevant mobility indicator as a control.

The two margins diverge. Sectoral transitions respond strongly and persistently while geographic mobility does not. Pre-recession sectoral coefficients (2003–2005) are near zero (Panel a), confirming parallel pre-shock trajectories. The sectoral response emerges in 2008 and peaks in 2009, when the interquartile shock difference raises the annual transition probability by roughly 5.0 percentage points, and coefficients stay large and precise through 2015 (3.8 to 4.2 points at the interquartile difference), reflecting continued reallocation out of a sector whose employment share remained permanently depressed.

Geographic mobility does not respond in economically or statistically meaningful ways over 2007–2015 (Panel b). Point estimates of the interquartile effect never exceed 0.5 percentage points, corresponding to less than 4 percent of the roughly 12 percent pre-recession baseline migration rate, and are statistically indistinguishable from zero throughout the post-recession period. Pre-trends are flat. Coefficients for 2004–2006 are small and not statistically different from zero, consistent with parallel trends in geographic mobility. The 2003 coefficient is negative and marginally significant (-0.091 , $p < 0.10$), suggesting slightly lower pre-recession mobility in more-exposed provinces, although the magnitude is small relative to post-recession sectoral adjustments.

A potential concern is that correlated shocks across connected labor markets may attenuate measured out-migration by depressing employment opportunities in destination provinces. To address this, I construct a destination-adjusted exposure measure that nets out such correlated demand shocks using pre-recession bilateral migration flows (Bertoli and

Moraga, 2013; Borusyak et al., 2022).

$$\text{Shock}_r^{\text{adj}} = \text{Shock}_r - \sum_{k \neq r} \mu_{r \rightarrow k} \text{Shock}_k. \quad (6)$$

Table 5 quantifies both margins at the cross-sectional level. At the interquartile shock difference, sectoral exit rises by roughly 3.9 percentage points (Columns 4–6), whereas interprovincial migration is essentially unchanged—under 0.3 percentage points, or 6 percent of the pre-recession baseline, and statistically indistinguishable from zero (Columns 1–3). Both results hold under the destination-adjusted shock of equation (6) (Columns 3 and 6). Adjustment to the collapse occurred almost entirely through movement out of construction into other sectors, with no meaningful geographic response.

Table 5: Geographic versus Sectoral Reallocation

	(1)	(2)	(3)	(4)	(5)	(6)
	Changed province			Changed sector		
Shock	−0.006 (0.120)	0.057 (0.091)		0.374*** (0.055)	0.377*** (0.054)	
Weighted shock			−0.047 (0.166)			0.288*** (0.079)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: The dependent variable in Columns (1)–(3) is an indicator for residing in a province different from the 2007 province of residence, measured in 2012. Columns (4)–(6) use an indicator for employment outside construction in 2012. Columns (2) and (5) additionally control for pre-recession average mobility, and Columns (3) and (6) control for the pre-migration adjusted shock. The adjusted shock nets out construction employment declines in connected provinces, weighted by pre-recession bilateral migration flows following equation (6). All specifications include the preferred set of individual and regional controls from Column (3) of Table 1. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

If adjustment runs mainly through sectoral and not geographic reallocation, the index should strongly capture who exits construction. The shock should accelerate departure for workers whose skills map onto available local demand and leave exit rates unchanged for those with weaker alignment. Table 6 shows this pattern. The shock raises the exit probability by $0.520 \times \text{Shock}$ for top-quartile workers against $0.283 \times \text{Shock}$ at the bottom; at the mean Q1 shock of 0.574, this is roughly 30 versus 16 percentage points.²³ Workers with the weakest skill–demand alignment show no greater propensity to leave construction

²³Table OA.3 in the Appendix uses as the dependent variable the probability of employment outside construction in 2012, showing an even more striking difference between workers in the upper quartiles of the reallocation index and those at the bottom.

as local conditions deteriorate, and these are the same workers who bear the largest and most persistent non-employment costs of Section 6.1.

Table 6: Sectoral Composition and the Probability of Leaving Construction

	(1)	(2)	(3)
	Sector Change, 2008–2012		
Shock	0.374*** (0.055)	0.376*** (0.051)	
Reall. Index		−0.032** (0.015)	
Shock × Reall. Index		0.074*** (0.024)	
<i>Shock × Reallocation Index Quartile:</i>			
Q1 (lowest)			0.283*** (0.059)
Q2			0.282*** (0.071)
Q3			0.413*** (0.071)
Q4 (highest)			0.520*** (0.056)
Controls	34,000 ✓	34,000 ✓	34,000 ✓

Notes: Linear probability estimates. Dependent variable: indicator for employment in a sector other than construction at any point between 2008 and 2012. All specifications include the preferred set of individual and regional controls from Column (3) of Table 1. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure B1 traces these dynamics. Pre-recession trajectories are parallel across quartiles, supporting the identifying assumption of no differential pre-trends. Following the shock, workers in the top quartile attain higher and more persistent non-construction employment than those in the bottom two quartiles, with little evidence of convergence over the subsequent seven years. Bottom-quartile workers exhibit a brief employment advantage in 2008, but this reverses by 2009 and widens thereafter. Reallocation destinations also differ systematically in the quality of jobs (Table OA.4). Bottom-quartile workers who leave construction disproportionately sort into hospitality and agriculture—cyclical sectors characterized by less stable employment—whereas top-quartile workers reallocate into sectors with more durable employment opportunities. For low-index workers, this combination of persistent employment gaps and systematically weaker destinations points to a binding reallocation constraint. Limited occupational overlap with local demand restricts access to viable alternatives, leaving the extensive margin of adjustment largely unavailable.

6.3 Index Decomposition

Whether skill alignment governs the magnitude of earnings losses depends on the underlying channel: the match between worker characteristics and local sectoral composition, the wage premia of locally available sectors, or a combination of both. I distinguish these mechanisms

using three measures. The baseline reallocation index weights skill–demand alignment by each sector’s median wage relative to construction. Removing the wage weights yields a pure skill–sector overlap measure that isolates the job-matching component. A third measure captures the local sectoral wage mix, constructed independently of skill alignment, and isolates whether a favorable wage distribution alone can account for the results. All three variables are standardized, so coefficients are expressed as the effect of a one-standard-deviation change.

The no-wage variant of the index retains strong predictive power (0.307), while the wage-mix variant flips sign (−0.482 in Column 3, −1.121 in Column 6), indicating that the predictive content derives from alignment between worker skills and the composition of surviving local demand rather than from differential local access to high-wage sectors (Table 7). Workers in provinces with higher-paying surviving industries do not recover faster; if anything, conditional on skill alignment, they fare worse. The binding constraint on post-displacement earnings recovery is occupational mismatch, not the wage level of available destinations. This finding echoes [Autor and Dorn \(2013\)](#) and [Lise and Postel-Vinay \(2020\)](#), who document persistent reallocation penalties for workers whose skills transfer poorly across sectors.

Table 7: Reallocation Index Variants

	Employment			Cumulative Earnings		
	Baseline (1)	No Wage (2)	Wage-mix (3)	Baseline (4)	No Wage (5)	Wage-mix (6)
Shock	−1.617*** (0.156)	−1.656*** (0.165)	−1.645*** (0.143)	−2.045*** (0.277)	−2.115*** (0.294)	−2.172*** (0.271)
Shock × Index	0.347*** (0.083)	0.307*** (0.073)	−0.482*** (0.155)	0.353** (0.160)	0.225 (0.155)	−1.121*** (0.364)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: Each column reports OLS estimates of the construction shock interacted with one variant of the reallocation index. Columns (1) and (4) use the baseline index $\sum_j \hat{\phi}_j(X_i) \times \hat{s}_{jr}/\bar{s}_j \times \widetilde{W}_j$, which combines skill–demand alignment, local vacancy intensity, and the relative wage weight. Columns (2) and (5) drop the relative wage ratio $\widetilde{W}_j$, isolating the skill–demand alignment component. Columns (3) and (6) replace worker-specific transition probabilities $\hat{\phi}_j(X_i)$ with uniform weights $1/J$, so that the index captures only the local sectoral wage mix independently of skill alignment. All index variants are standardised to mean zero and unit standard deviation. Level terms for each index are included in all specifications but not reported. All regressions include the individual controls in the baseline specification of Table 1. Standard errors clustered at the province level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Match is non-separable across levels. A worker’s alignment reflects both the opportunity set common to their province and their position relative to observationally similar workers within it. Treating the index as a single regional component might leave the second source of

variation unexploited. I therefore decompose it into a province mean, capturing the average local opportunity set, and a within-province residual, capturing alignment differences among observationally similar workers in the same province.

Table 8: Regional versus Worker-Level Reallocation Index

	Employment			Cumulative Earnings		
	(1) Regional	(2) Worker	(3) Joint	(4) Regional	(5) Worker	(6) Joint
Shock	−1.569*** (0.145)	−1.616*** (0.156)	−1.569*** (0.145)	−1.976*** (0.247)	−2.045*** (0.278)	−1.977*** (0.249)
Shock × Regional	0.144 (0.133)		0.137 (0.134)	0.457* (0.233)		0.441* (0.234)
Shock × Worker-level		0.332*** (0.080)			0.344** (0.162)	
Shock × Within			0.296*** (0.063)			0.201 (0.135)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: Dependent variables are cumulative years of employment (columns 1–3) and cumulative normalized earnings (columns 4–6), 2007–2012. The reallocation index measures the expected value of transitioning out of construction, standardized to mean zero and unit variance. The regional index is the province mean of the reallocation index, standardized to unit variance; the within-province index is the individual deviation from the province mean, standardized to unit variance. Both components enter columns (3) and (6) simultaneously. The regional and within-province components are constructed as orthogonal on the raw scale. All specifications include individual, contract, and regional controls. Clustered standard errors at the province level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8 reports the two components separately and jointly. For cumulative days employed the within-province component dominates. In the joint specification the shock interaction with the worker-level component is 0.296, roughly 2.2 times the regional component (0.137). The regional interaction loses precision when both enter jointly, but its magnitude, nearly half the within-province estimate, indicates that provincial structure contributes to employment recovery even where the available variation cannot identify it separately. Reemployment speed thus depends primarily on how well a worker’s skills match locally available sectors, with provincial structure secondary but non-negligible.

The pattern reverses for cumulative earnings, with a near-identical ratio in the opposite direction. Provincial structure, through the sectoral presence, size, and wages a worker can land in, therefore accounts for most of the shock-driven variation in earnings losses, with worker-level alignment accounting for the remainder. This parallels Yi et al. (2024), who show that local industry composition moderates earnings losses from trade shocks; the decomposition here complements that evidence by recovering the worker-level alignment component within the region-industry margin.

The opposing 2:1 ratios confirm the same partition. Employment is governed by worker-level occupational compatibility, while earnings additionally reflect sectoral wage premia across destinations—so reallocation capacity operates mainly through the speed of reemployment, with limited effects on post-reemployment wages.

Place-based strategies targeting high-wage industries rely on agglomeration-driven spillovers in employment (Moretti, 2011), while diversified local economies are expected to facilitate adjustment through multiple reallocation channels (Jacobs, 1984). The evidence here suggests these mechanisms are strongly conditional on worker–sector compatibility. Reallocation occurs only when worker skills align with local labor demand, and adjustment costs increase with occupational distance (Gathmann and Schönberg, 2010). During the Great Recession, displaced construction workers did not move systematically into expanding sectors such as finance, healthcare, or professional services because occupational mismatch constrained feasible transitions. This implies that agglomeration and diversity are not sufficient for adjustment; the relevant margin is the ability to traverse occupational boundaries.

Table OA.9 in the Online Appendix compares RI^w against a Caldwell and Danieli (2024) style outside-options index (OOI^{CD}) constructed from the pre-shock sectoral distribution, adapted to use province-level rather than national sectoral shares.²⁴ Estimated alone, OOI^{CD} is statistically indistinguishable from zero for employment recovery (Column 2), and the result is unchanged when construction is retained in the opportunity set (Column 3). In the joint specification, the RI^w interaction rises to 0.383 while OOI^{CD} remains negative and imprecise (Column 4). This pattern reflects a structural difference between the two measures. OOI^{CD} summarizes the pre-shock distribution of a worker’s sectoral opportunities—a portable worker attribute suited to wage bargaining in steady state—while RI^w conditions on post-shock surviving demand within the worker’s province and captures variation in offer arrival rates. When the origin sector effectively ceases hiring, what matters for reemployment speed is the alignment between worker skills and the sectors that continue to post vacancies locally—variation the pre-shock stock distribution cannot recover.

7 Policy Counterfactual

Improving a worker’s index position is analogous to shifting the sector-matching function upward—the margin occupational retraining and targeted job-search assistance are designed to improve. Using the quartile coefficients of Table 4, I ask how cumulative earnings losses

²⁴The original index is designed for steady-state wage bargaining and exploits commuting distance to identify variation in outside options; neither ingredient is available or appropriate here. The comparison serves to illustrate why a pre-shock stock distribution cannot account for reemployment speed following a large sectoral collapse.

would change if active labor market programs raised the reallocation capacity of the worst-aligned workers. The exercise is an accounting calculation, not a program evaluation. It holds fixed the provincial shock, the post-recession demand composition, and untreated workers' outcomes, and assumes that shifting workers toward higher-index states does not crowd out others or compress wages in receiving sectors. It applies to programs that accelerate sectoral reentry rather than to income support, and should be read as an upper bound on what reallocation-targeted training could recover.

Spanish ALMP during the Great Recession was not conditioned on reallocation capacity, leaving this predictable heterogeneity unexploited. The bottom and second index quartiles faced cumulative earnings-loss coefficients of -2.462 and -2.039 .

Define worker i 's loss in quartile k as $L_i = \hat{\beta}_k \times \text{Shock}_{r_i}$. Targeting moves treated Q_1 workers to a higher-quartile coefficient while workers in Q_2 through Q_4 remain at actual outcomes, giving an aggregate per-worker recovery of

$$\Delta^{\text{CF1}} = \frac{N_{Q_1}}{N} \cdot (\hat{\beta}_{Q_1} - \hat{\beta}_{Q_2}) \cdot \overline{\text{Shock}}, \quad (7)$$

where $\overline{\text{Shock}} = 0.574$ is the mean shock among Q_1 workers and $N_{Q_1}/N = 0.25$ by construction.

Moving Q_1 workers to Q_2 alignment recovers about 17 percent of the treated group's earnings loss, or roughly 5 percent of the full-sample average once scaled by the Q_1 share. A more intensive shift to the Q_4 distribution—an upper bound on skill-reorientation programs—recovers about a third of the treated group's loss and roughly 10 percent of aggregate losses; because recovery and baseline loss scale with the same shock, these shares are invariant to the shock level.²⁵

The intensive-scenario gain is non-negligible against typical European ALMP returns (Card et al., 2018), though it still leaves most of the bottom-quartile loss intact. Targeting is nonetheless feasible ex ante. The index is constructible from pre-displacement records on education, occupational skill group, and province, and captures adjustment heterogeneity that no single observable accounts for (Table OA.8).

²⁵Per-unit-shock recovery is $\hat{\beta}_{Q_1} - \hat{\beta}_{Q_2} = 0.423$ (to Q_2) and $\hat{\beta}_{Q_1} - \hat{\beta}_{Q_4} = 0.872$ (to Q_4), i.e. $0.423/2.462 \approx 17$ and $0.872/2.462 \approx 35$ percent of the treated group's loss; weighting by $N_{Q_1}/N = 0.25$ gives aggregate shares of about 5 and 10 percent of the full-sample average loss ($2.067 \times 0.574 \approx 1.186$).

8 Additional Robustness Checks

Sample composition. The baseline results are insensitive to the age ceiling and the restriction to native workers (Table C5). Extending the sample to workers aged up to 55 and additionally incorporating foreign-born workers leaves both the earnings and employment coefficients within ten percent of the baseline estimates, despite immigrants’ well-documented tendency to respond to local demand shocks through geographic relocation (Cadena and Kovak, 2016).

Minimum attachment threshold. The earnings and employment coefficients are stable across minimum-attachment thresholds of three, six, nine, and twelve months, spanning a 22 percent variation in sample size (Table C6). The stability at the three-month threshold—which admits workers with weaker pre-recession attachment—addresses the concern that stricter cutoffs select on workers whose 2007 employment was itself a product of the boom.

Instrumenting the shock. Instrumenting the provincial shock and its interaction with the index leaves the central result intact. The shock–index interaction rises from 0.332 to 0.570 for employment and from 0.344 to 0.703 for earnings (Table C3). Two forces likely account for the amplification. Classical measurement error in the provincial shock attenuates the OLS interaction toward zero. The IV estimates also reflect a local average treatment effect concentrated in provinces where pre-recession construction growth was most pronounced, which tend to be those where the collapse was largest. At a 75th-percentile shock, a one-standard-deviation improvement in alignment offsets about one-fifth of losses under OLS and one-third under IV. First-stage diagnostics support this reading. Sanderson–Windmeijer conditional F-statistics are 12.51 for the shock and 39.96 for the interaction, and the Anderson–Rubin test rejects jointly zero coefficients ($F = 11.84$, Table OA.2).

Inference robustness. Table C4 reports wild bootstrap p-values from 1,000 replications with Rademacher weights, clustering by province. The shock is significant at the one percent level for cumulative employment and earnings; the p-value of 0.063 for average wage reflects the noisier character of this outcome throughout the analysis. The Shock $\times$ RI^w interaction is significant at the one percent level for cumulative employment ($p = 0.000$) and at the ten percent level for cumulative earnings ($p = 0.071$), and is not significant for average wage. The finite-sample correction modestly widens inference for the earnings interaction while leaving the employment result unchanged.

Generated-regressor correction. Because $\widehat{RIw}_i$ is constructed from a first-stage multinomial logit estimated on a separate training sample, the second-stage standard errors do not account for sampling variation in $\hat{\varphi}_j(X_i)$, raising the possibility of generated-regressor bias (Pagan, 1984). I address this via a province-level block bootstrap that propagates un-

certainty through both estimation stages.²⁶ Table OA.10 reports the results. Bootstrap and province-clustered OLS standard errors are within 10 percent of each other across all three outcomes (employment: 0.085 vs. 0.080; cumulative earnings: 0.153 vs. 0.162; conditional wage: 0.003 vs. 0.003), indicating that first-stage sampling variation contributes negligibly to second-stage uncertainty. Bootstrap percentile intervals exclude zero for the employment interaction at both the 90 and 95 percent level ([0.216, 0.486] and [0.185, 0.521]) and for cumulative earnings at conventional levels ([0.073, 0.579] and [0.004, 0.644]). The conditional wage interval is centered near zero ([−0.003, 0.006] at 90%), consistent with the results in the main text.

Province fixed effects. Table C8 adds province fixed effects, absorbing the shock’s main effect and all time-invariant provincial confounds, to confirm that the $\text{Shock} \times RI^w$ interaction is identified from within-province cross-worker variation. The employment interaction is essentially unchanged at 0.322 (SE = 0.070, $t = 4.60$), a 3 percent attenuation relative to the baseline 0.332. The earnings interaction attenuates to 0.219 and loses precision, consistent with the decomposition in Table 8. Cumulative earnings reflect both reemployment speed—a within-province worker-level object—and the sectoral wage structure of reemployment destinations, which varies primarily across rather than within provinces. Since the model’s central prediction concerns offer arrival rates, the employment result is the primary mechanism test, and it is essentially unchanged.

Reallocation index and worker and regional characteristics. A potential concern is that the reallocation index captures variation in observable worker characteristics—education, occupational skill, labor market experience—or provincial conditions rather than skill–demand alignment per se. Table C7 addresses this by augmenting the baseline interaction with shock interactions for each characteristic separately. The $\text{Shock} \times RI^w$ coefficient is stable across columns (1)–(6). It ranges from 0.267 to 0.332, a maximum attenuation of 20 percent relative to the baseline, and remains significant at conventional levels throughout. The HHI and unemployment rate interactions (columns 5–6) leave the coefficient essentially unchanged at 0.308, indicating the index does not simply proxy for local labor market diversity or pre-recession slack. Table OA.6 provides a complementary test by replacing the continuous index with decile fixed effects, flexibly absorbing any level relationship between

²⁶The procedure resamples the 50 provinces with replacement: for each replication b , I (i) draw a bootstrap sample of provinces; (ii) re-estimate the multinomial logit on the resampled training sample (non-construction hiring spells, 2002–2006); (iii) apply the replication- b coefficients to the resampled estimation sample to obtain $\hat{\varphi}_j^b(X_i)$ and recompute $\widehat{RIw}_i^b$; and (iv) run the second-stage OLS on the resampled outcome sample, standardizing $\widehat{RIw}_i^b$ using the fixed baseline mean and standard deviation to hold the unit of measurement constant across replications. Within-replication standard errors cluster on the bootstrap province identifier. Bootstrap standard errors are the standard deviation of each coefficient’s empirical distribution across 1,000 replications.

the index and outcomes. The employment interaction remains stable at 0.232, consistent with the result not being driven by a monotone association between index level and general worker ability. The earnings interaction declines to 0.211 and loses precision, consistent with the province FE result in Table C8. Earnings losses additionally reflect the sectoral wage structure of local demand, which is partially collinear with the index level and varies mainly across rather than within provinces.

Province-specific selection in the first stage. A potential concern is that the multinomial logit transition probabilities absorb province-specific selection patterns, so that the index partly reflects local unobservables correlated with resilience rather than skill–demand alignment per se. To address this, I re-estimate the index using a leave-one-province-out procedure. For workers in province p , the transition probabilities are estimated on workers from all other provinces and applied to workers in p , ensuring the first stage cannot encode province-specific selection. The leave-out index has a correlation of 0.964 with the baseline, indicating that the multinomial logit coefficients are stable across provinces. The interaction coefficients are essentially unchanged. The employment interaction moves from 0.332 to 0.334 and the earnings interaction from 0.344 to 0.339, with the conditional-wage null remaining at 0.002 throughout (Table C10).

Reallocation dynamics. Year-by-year estimates of the shock interacted with bottom- and top-quartile index indicators show no differential pre-trends. Q1 and Q4 coefficients are small and statistically indistinguishable from zero throughout 2003–2007 on both employment and sectoral exit, so the index does not proxy pre-existing differences in worker outcomes (Figure B1). The Q1–Q4 gap opens immediately after the shock, deepens through $t = 3$ – 4 , and persists through $t = 8$ without convergence, inconsistent with a transitory interpretation of the moderating effect.

Reemployment hazard. Table C9 estimates cause-specific reemployment hazards separately for workers who transition immediately following displacement and for those who experience a non-employment spell, providing a direct test of Proposition 1(i). Among workers who move directly into another job, the hazard ratio for the highest alignment quartile is 1.906 relative to otherwise similar workers exposed to the same shock. Among workers who enter non-employment, the corresponding hazard ratio is 3.286.²⁷ Both estimates support the proposition that better-aligned workers reallocate more rapidly following displacement.

²⁷The Q1–Q4 contrast is estimated imprecisely among non-employment spells, reflecting the relatively small number of observed failures in this subsample, but point estimates are consistently positive and economically large across specifications.

9 Conclusion

I have studied how local sectoral composition shapes worker adjustment to large, persistent sectoral shocks. Using Spain’s construction collapse—a contraction of roughly 60 percent that was spatially uneven—I document three findings consistent with the alignment mechanism. First, displacement costs operated predominantly through extended non-employment and not through lower wages. Second, adjustment ran through intersectoral, not geographic, mobility. Third, workers whose skill profiles aligned with locally available sectors exited non-employment faster, and experienced smaller earnings losses, though the earnings dimension depends additionally on the sectoral wage structure of local demand.

To do so I develop a worker-level reallocation index that formalizes the mechanism as an offer-arrival rate, and that predicts reemployment speed but not the conditional wage upon matching. This partition separates the alignment mechanism from match-quality and wage-setting alternatives.

The mechanism is not specific to Spain. Its logic requires only that geographic mobility be incomplete in the short and medium run and that local economies differ in sectoral composition—conditions that characterize trade-exposed manufacturing regions (Autor et al., 2014; Dix-Carneiro and Kovak, 2017) and labor markets reshaped by automation (Acemoglu and Restrepo, 2020). The index is constructible from matched worker–firm data and a pre-shock window, inputs increasingly available across OECD countries. Estimated magnitudes are specific to Spain; the structure is not.

The results carry direct implications for adjustment policy. Standard programs implicitly treat all workers within a location as facing identical adjustment constraints. Within-market heterogeneity in reallocation capacity is large and predictable ex ante from administrative records, suggesting that directing training and job-search assistance toward workers poorly aligned with locally growing sectors would improve on current targeting. The equilibrium effects of such interventions on local sectoral composition—and hence on future reallocation capacity—remain an open question.

A Reallocation Index: Theoretical Framework

This appendix develops the search model that motivates the reallocation index of Section 4. Job seekers allocate search effort across local sectors, firms post vacancies subject to congestion, and local sectoral composition is fixed by predetermined production technology. The model rationalizes the index’s functional form, delivers the predictions tested in the body, and establishes that the index equals the model-implied rate at which a worker draws

acceptable offers.

A.1 Environment

Time is continuous. The economy has R provinces indexed by r , each with J sectors indexed by j , including construction (cs). At $t = 0$, a unit mass of workers in province r is displaced from construction by a permanent contraction in construction demand, which I assume is unanticipated. Workers do not relocate across provinces, consistent with Section 6.2.²⁸

Worker i has observable skills $X_i \in \mathcal{X}$ determining productivity across sectors:

$$y_j(X_i) = a_j \cdot \phi_j(X_i), \quad (8)$$

where $a_j > 0$ is sector-specific TFP and $\phi_j(X_i) \in [0, 1]$ is match quality, interpreted as the probability that the worker's skills meet the sector's task requirements.

Sector j in province r posts vacancies v_{jr} under free entry. Matching follows a Cobb-Douglas technology with sector-specific congestion, giving an offer-receipt rate for a worker searching in sector j of

$$m_{jr} = m_0 \cdot u_{jr}^{1-\eta} \cdot v_{jr}^\eta, \quad (9)$$

$$p_{jr} = \frac{m_{jr}}{u_{jr}} = m_0 \cdot \theta_{jr}^\eta, \quad (10)$$

where u_{jr} is the measure of workers searching in sector j of province r , $\eta \in (0, 1)$ the vacancy elasticity, $m_0 > 0$ baseline matching efficiency, and $\theta_{jr} = v_{jr}/u_{jr}$ tightness.

Let $\sigma_j(X_i, r) \in [0, 1]$ denote the share of search effort worker i directs to sector j . The flow rate of acceptable offers from sector j is

$$\lambda_{ij}(r) = \sigma_j(X_i, r) \cdot p_{jr} \cdot \mathbb{P}(\text{accept} \mid \text{offer}, j, X_i), \quad (11)$$

and the worker accepts if $W_j(X_i) \geq U(X_i, r)$, where W_j and U are the values of employment in sector j and of unemployment. Unemployed workers receive flow utility b ; employed workers earn a Nash-bargained wage with worker power $\beta \in (0, 1)$, giving employment value

$$w_{jr}(X_i) = \beta \cdot y_j(X_i) + (1 - \beta) \cdot b, \quad (12)$$

$$\rho W_j(X_i) = w_{jr}(X_i) + \delta [U(X_i, r) - W_j(X_i)]. \quad (13)$$

²⁸The no-migration assumption matches the empirical pattern; it is not a structural prediction of the model.

Firms pay flow cost κ_j per vacancy; free entry requires $\kappa_j = q_{jr} \cdot J_j(X_i)$, where $q_{jr} = m_0 \cdot \theta_{jr}^{\eta-1}$ is the vacancy-filling rate and $J_j(X_i)$ the firm's value of a filled position.

A.2 Equilibrium Search Allocation

The worker chooses $\{\sigma_j(X_i, r)\}_{j \neq cs}$ to maximize the value of unemployment:²⁹

$$\rho U(X_i, r) = b + \max_{\{\sigma_j\}} \sum_{j \neq cs} \sigma_j(X_i, r) \cdot p_{jr} \cdot \max\{W_j(X_i) - U(X_i, r), 0\}. \quad (14)$$

The objective is linear in $\{\sigma_j\}$, so without search costs the optimum concentrates all effort on the highest-return sector. A quadratic search cost $\chi(\sigma_j) = \sigma_j^2/2$ delivers interior allocations and a closed form.³⁰ The first-order condition gives

$$\sigma_j^*(X_i, r) = p_{jr} \cdot S_j(X_i), \quad (15)$$

where $S_j(X_i) = W_j(X_i) - U(X_i, r)$.

A.3 The Reallocation Index

Three functional-form restrictions, adopted for tractability, reduce the model to the index of Section 4 (derivations in Online Appendix OA.A.4).

Restrictions A1–A2 (linear surplus; tightness reflects local specialization).

Linearizing surplus around average match quality and combining free entry with matching (9) give

$$S_j(X_i) \approx \frac{\beta}{\rho + \delta} \cdot a_j \cdot \phi_j(X_i), \quad (16)$$

$$\theta_{jr}^* = \left(\frac{m_0 \cdot \bar{S}_j}{\kappa_j} \right)^{1/(1-\eta)} \cdot \frac{s_{jr}}{\bar{s}_j}, \quad (17)$$

where $\bar{S}_j$ is average sector- j surplus and $s_{jr}/\bar{s}_j$ local specialization. A1 is a first-order approximation obtained by substituting (12) and (8) into the value function; A2 follows because sectors with greater local specialization attract proportionally more vacancy posting.

Restriction A3 (acceptance scales with relative wages). The share of received offers accepted is proportional to w_j/w_{cs} , from aggregating acceptance decisions over reser-

²⁹Search is restricted to non-construction sectors, reflecting the permanent contraction in local construction demand.

³⁰Sectors enter separably, so the first-order condition needs no Lagrange multiplier and the σ_j^* need not sum to one; the index is standardized in the empirical implementation.

vation wages spanning the sectoral wage distribution.

Reallocation index. The flow rate of acceptable offers from sector j is $\lambda_{ij}(r) = \sigma_j^*(X_i, r) \cdot p_{jr} \cdot \alpha_j(X_i)$. Summing across sectors, substituting A1–A3, and normalizing by Λ_0 —which absorbs m_0 , β , ρ , δ , κ_j , and the search-cost parameter, and is common across workers and provinces—recovers the index of equation (1):³¹

$$\Lambda_i(r) = \Lambda_0 \cdot \sum_{j \neq cs} \phi_j(X_i) \cdot \frac{s_{jr}}{\bar{s}_j} \cdot \frac{w_j}{w_{cs}}, \quad (18)$$

$$\text{RI}^w(X_i, r) \equiv \frac{\Lambda_i(r)}{\Lambda_0} = \sum_{j \neq cs} \phi_j(X_i) \cdot \frac{s_{jr}}{\bar{s}_j} \cdot \frac{w_j}{w_{cs}}. \quad (19)$$

The three components map directly to the frictions identified in Section 4: skill compatibility, local vacancy intensity, and acceptance probability. They enter multiplicatively because, under A1–A3, these forces are non-separable determinants of the offer arrival rate.

Destination distribution. The probability that worker i reemploys in sector j is proportional to the acceptable-offer rate from that sector:

$$\pi_j(X_i, r) = \frac{\phi_j(X_i) \cdot (s_{jr}/\bar{s}_j) \cdot (w_j/w_{cs})}{\sum_{k \neq cs} \phi_k(X_i) \cdot (s_{kr}/\bar{s}_k) \cdot (w_k/w_{cs})}. \quad (20)$$

Two workers with identical RI^w share the same reemployment hazard but may differ in destination composition, one concentrating reemployment in a few well-matched sectors and another spreading it across more locally available options.

A.3.1 Testable Predictions

Proposition 1 (Predictions). *Under the environment of Sections A.1–A.3 and Restrictions A1–A3: (i) expected non-employment duration is strictly decreasing in $\text{RI}^w(X_i, r)$; (ii) the sectoral transition hazard is proportional to $\text{RI}^w(X_i, r)$; (iii) expected cumulative earnings losses over any post-displacement horizon are decreasing in $\text{RI}^w(X_i, r)$.*

Proof. (i) Conditional on $\{\sigma_j^*\}$ and $\{\theta_{jr}^*\}$, acceptable offers arrive as a Poisson process with rate $\Lambda_i(r) = \Lambda_0 \cdot \text{RI}^w(X_i, r)$, so expected non-employment duration is $1/[\Lambda_0 \cdot \text{RI}^w(X_i, r)]$, decreasing in the index. (ii) The sectoral transition hazard equals $\Lambda_i(r)$ by construction,

³¹Full derivations are in Online Appendix OA.A.4. The reduction is not exact: province-invariant, sector-specific constants $\bar{B}_j \cdot A_j$ are absorbed into the alternative-specific intercepts of the multinomial logit used to estimate $\phi_j(X_i)$.

depending on (X_i, r) only through the index. (iii) Cumulative losses combine foregone earnings during non-employment—governed by the index as in (i)—and the reemployment wage gap. The expected reemployment wage $\sum_j \pi_j(X_i, r) w_j$ depends on (X_i, r) through the index components via (20). The wage component is approximate. The derivation assumes reservation wages are distributed around w_{cs} , and the approximation is tighter the smaller the dispersion of reservation wages across workers. $\square$

A.4 Extensions of the Baseline Framework

The baseline index corresponds to the equilibrium offer arrival rate under two restrictions calibrated to the Spanish construction collapse: limited geographic mobility ($\omega = 1$) and the effective disappearance of construction as a source of new employment ($\gamma = 0$), both supported empirically in Sections 2 and 6.2. Each can be relaxed while preserving the index’s modular structure.

When workers can search in both the home province r and a neighboring province $\tilde{r}$, the index extends to:

$$\text{RI}^w(X_i, r, \tilde{r}) = \sum_{j \neq cs} \phi_j(X_i) \cdot \frac{\omega \cdot s_{jr} + (1 - \omega) \cdot s_{j\tilde{r}}}{\bar{s}_j} \cdot \frac{w_j}{w_{cs}}, \quad (21)$$

where $\omega \in [0, 1]$ weights the home province and $\omega = 1$ recovers the baseline. Geographic mobility enters exclusively through the local opportunity component $s_{jr}/\bar{s}_j$; the skill–demand match $\phi_j(X_i)$ and wage-acceptance term w_j/w_{cs} are unaffected. Geographic access and skill–demand alignment are therefore substitutes. Workers with low home-province alignment can partially compensate by drawing on a better-matched neighboring market. The parameter ω is identified by cross-regional variation in bilateral mobility costs and could be estimated in settings where geographic adjustment is quantitatively important, trade-exposed commuting zones being a natural application.

When the contraction is incomplete, construction retains a fraction $\gamma \in [0, 1)$ of its pre-shock vacancy rate, and the index becomes:

$$\text{RI}^w(X_i, r, \gamma) = \sum_j \phi_j(X_i) \cdot \frac{s_{jr}(\gamma)}{\bar{s}_j} \cdot \frac{w_j}{w_{cs}}, \quad (22)$$

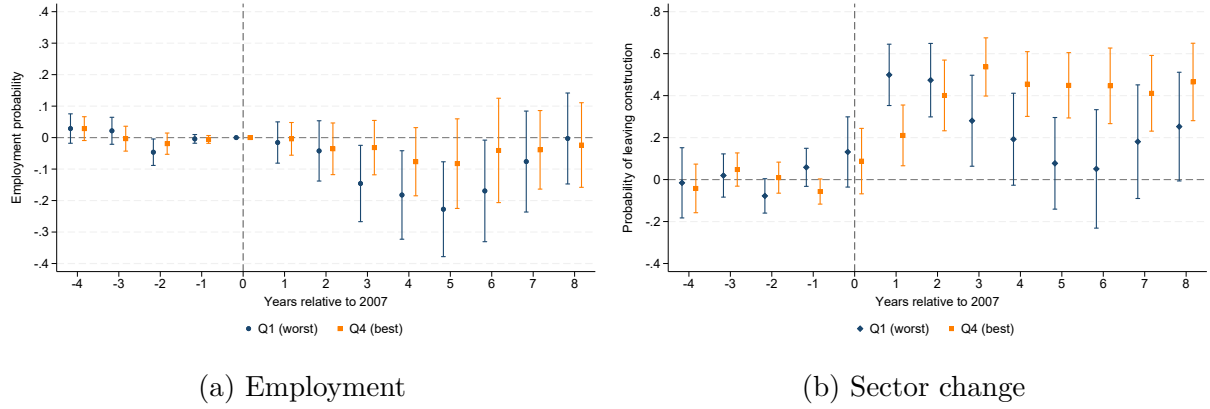
where $s_{cs,r}(\gamma) = \gamma \cdot s_{cs,r}$ and the summation includes construction; $\gamma = 0$ recovers the baseline. As γ rises, workers with high pre-shock construction attachment experience a smaller decline in effective search opportunities, and the index converges toward a conventional outside-options measure. The $\gamma \approx 0$ restriction is an implication of the free-entry condition: a

permanent contraction implies a proportional decline in vacancy posting as expected profits fall below κ_j .

Unlike the outside-options index of [Caldwell and Danieli \(2024\)](#), which is a portable worker attribute, the reallocation index is a worker-in-province object. Two workers with identical X_i face different prospects when local sectoral compositions differ (Section 6). The partial-shutdown extension is most relevant for gradual sectoral transitions—energy, automation—where γ can be proxied by the sector’s vacancy rate trajectory and identified from cross-worker variation in $\phi_{cs}(X_i)$ conditional on the aggregate shock.

B Additional Figures

Figure B1: Probability of Employment and Sectoral Exit by Reallocation Index Quartile

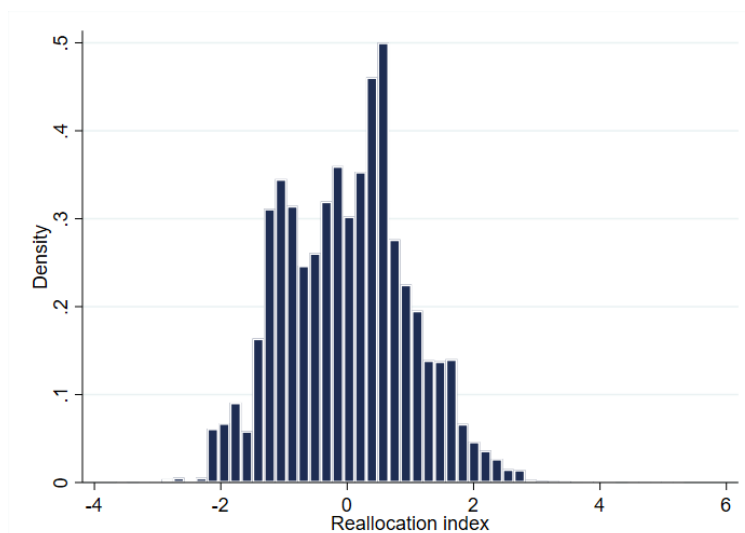


(a) Employment

(b) Sector change

Notes: Year-by-year linear probability estimates. Dependent variable: indicator for employment and for employment in a sector other than construction in year t , for $t \in \{2003, \dots, 2015\}$. Each year is estimated separately as a regression of the annual sector-change indicator on the shock interacted with indicators for each quartile of the reallocation index, conditional on quartile fixed effects and the pre-shock (2003–2006) average of the sector-change indicator. Controls and sample restrictions follow Table 1. Standard errors clustered at the province level.

Figure B2: Distribution of the Reallocation Index



Notes: Distribution of the normalized reallocation index among workers employed in the construction sector in 2007. The measure is standardized to have mean zero and unit standard deviation in the estimation sample.

C Additional Tables

Table C1: Workers' wage and employment trajectories, 2003-2006

	Cumulative earnings			Avg. Earn-ings	Employment
	(1)	(2)	(3)	(4)	(5)
Pre-recession construction growth	15.71 (33.16)	5.239 (30.65)	-131.3 (135.5)	-4.426 (4.521)	0.122 (0.330)
Observations	30,253	30,253	30,253	30,253	30,253
Age $\times$ Gender $\times$ Education	✓	✓	✓	✓	✓
Job characteristics		✓	✓	✓	✓
Regional controls			✓	✓	✓

Notes: The dependent variables are computed over the pre-recession period 2003–2006: cumulative earnings, average annual earnings, and cumulative days worked. The regressor of interest is the instrument—average annual construction sector growth over 2000–2006—which in the main specification instruments for the provincial employment shock. Control variables follow the Table 1. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C2: First stage, pre-recession construction growth on construction sector decline by province, 2007-2012

	(1)	(2)	(3)	(4)
	Construction sector decline, 2007-2012			
Pre-recession construction growth	1.287*** (0.365)	1.274*** (0.369)	1.290*** (0.367)	1.209*** (0.374)
Observations	34,000	34,000	34,000	34,000

Notes: The main regressor is the instrument—average annual growth in construction employment over 2000–2006—on the provincial employment shock, defined as the proportional change in the construction sector's employment share between 2007 and 2012. Control variables follow Table 1. Column (1) reports the baseline specification. Column (2) adds average provincial growth in mortgage values over 2003–2006. Column (3) instead includes average growth in the value of new mortgages over 2003–2006. Column (4) adds average provincial growth in the immigrant population over 2000–2006. Standard errors are clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C3: Robustness Reallocation Index and Earnings Losses: OLS and IV Estimates

	Employment		Cumulative Earnings	
	(1) OLS	(2) IV	(3) OLS	(4) IV
Shock	-1.616*** (0.156)	-2.212*** (0.461)	-2.045*** (0.278)	-3.180*** (0.747)
Shock $\times$ Reall. Index	0.332*** (0.080)	0.570*** (0.147)	0.344** (0.162)	0.703*** (0.264)
Observations	34,000	34,000	34,000	34,000

Notes: Dependent variable is cumulative normalised earnings, 2007–2012. The reallocation index measures the expected earnings value of transitioning out of construction, standardised to mean zero and unit variance. The shock is the province-level decline in construction employment share, 2007–2012. Columns (2) and (4) instrument the shock with average annual construction employment growth, 2000–2006. Column (4) additionally instruments the shock $\times$ index interaction with pre-recession growth $\times$ index. All specifications include individual, contract, and regional controls. Clustered standard errors at the province level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C4: Wild Bootstrap Inference: Reallocation Index Specifications

	(1) Employment	(2) Cumulative Earnings	(3) Avg. Earnings
<i>Panel A: Additive specification</i>			
Shock	0.000	0.000	0.063
Reallocation index	0.020	0.000	0.007
<i>Panel B: Interaction specification</i>			
Shock	0.000	0.000	0.063
Reallocation index	0.002	0.361	0.855
Shock $\times$ index	0.000	0.071	0.665

Notes: Wild bootstrap p -values from 1,000 replications with Rademacher weights, clustering by province. Null imposed throughout. The wild bootstrap addresses finite-sample distortions with 50 province clusters.

Table C5: Cumulative earnings and employment Impact of the Construction Employment Decline, extended age and nativity

	Cumulative earnings			Employment		
	(1)	(2)	(3)	(4)	(5)	(6)
Shock	-2.062*** (0.293)	-1.942*** (0.286)	-2.016*** (0.291)	-1.625*** (0.160)	-1.598*** (0.162)	-1.508*** (0.170)
Observations	34,000	37,796	47,011	34,000	37,796	47,011

Notes: Cumulative earnings and cumulative days employed effect of the collapse in construction-sector employment. This specifications extend the baseline analysis by including older and foreign-born workers. Columns (1) and (4) replicate the baseline specification. Columns (2) and (5) expand the sample to include workers up to age 55. Columns (3) and (6) further include foreign-born workers. All specifications include the preferred set of individual and regional controls from Column (3) of Table 1. Robust standard errors are clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C6: Robustness to pre-recession attachment thresholds

	(1) ≥ 3 months	(2) ≥ 9 months	(3) ≥ 12 months	(4) ≥ 3 months	(5) ≥ 9 months	(6) ≥ 12 months
	<i>Panel A. Earnings and employment outcomes</i>					
	Cumulative earnings			Employment		
	(1)	(2)	(3)	(4)	(5)	(6)
Shock	-2.067*** (0.259)	-2.026*** (0.307)	-2.032*** (0.312)	-1.614*** (0.138)	-1.694*** (0.168)	-1.699*** (0.183)
Observations	35,310	30,039	27,663	35,310	30,039	27,663
	<i>Panel B. Mobility outcomes</i>					
	Changed province			Changed sector		
Shock	-0.033 (0.116)	-0.011 (0.122)	0.008 (0.124)	0.352*** (0.050)	0.407*** (0.056)	0.453*** (0.056)
Observations	35,310	30,039	27,663	35,310	30,039	27,663

Notes: The table reports estimates of the effect of local shock exposure on cumulative outcomes over the post-2007 period under three alternative pre-recession attachment thresholds. Columns (1)–(3) and (4)–(6) successively require workers to have held employment for at least 3, 9, and 12 months in 2007. Panel A reports cumulative earnings and employment outcomes; Panel B reports the probability of changing province of residence and the probability of changing sector of employment. All specifications include the same control variables as in the main specification of Table 1. Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C7: Reallocation Index vs. Worker and Regional Characteristics

	Dependent variable: Employment					
	(1) Baseline	(2) + Educ	(3) + Skill	(4) + Exp	(5) + HHI	(6) + Unempl
Shock	-1.616*** (0.156)	-1.667*** (0.170)	-0.433 (0.585)	-2.211*** (0.213)	-1.143*** (0.419)	-1.234** (0.515)
Shock $\times$ RI	0.332*** (0.080)	0.317*** (0.103)	0.290* (0.158)	0.267*** (0.081)	0.308*** (0.076)	0.308*** (0.075)
Observations	34,000	34,000	34,000	34,000	34,000	34,000

Notes: All specifications replicate the baseline employment specification in Table 3. Columns (2)–(6) further allow the effect of the shock to vary with workers' education, skill group, potential experience, local labor market concentration (HHI), and the provincial unemployment rate in 2006 by interacting each characteristic with the shock measure. Standard errors are clustered at the province level and reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C8: Robustness to Province Fixed Effects and deciles Reall. Index

	Employment		Earnings	
	(1) Baseline	(2) Province FE	(3) Baseline	(4) Province FE
Shock	-1.616*** (0.156)		-2.045*** (0.278)	
Shock $\times$ RI^w	0.332*** (0.080)	0.322*** (0.070)	0.344** (0.162)	0.219 (0.146)
Observations	34,000	34,000	34,000	34,000
Province FE	No	Yes	No	Yes
Baseline controls	Yes	Yes	Yes	Yes

Notes: All specifications replicate the baseline employment specification in Table 3. Dependent variables are cumulative years of employment (columns 1–2) and normalized cumulative earnings (columns 3–4), both measured over 2007–2012. Province fixed effects in columns (2) and (4) absorb the shock main effect, which is therefore not identified and omitted. Standard errors are clustered at the province level and reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C9: Competing Risks: Immediate vs. Non-Immediate Reemployment

	Immediate reemployment			Non-immediate reemployment		
	(1)	(2)	(3)	(4)	(5)	(6)
RI^w	1.026*** (0.009)	0.818*** (0.041)		0.919*** (0.022)	0.670*** (0.101)	
Shock		1.160 (0.113)			1.708* (0.492)	
Shock $\times$ RI^w		1.487*** (0.127)			1.748** (0.422)	
Shock $\times$ Q1			0.891 (0.160)			1.232 (0.576)
Shock $\times$ Q2			1.018 (0.192)			0.547 (0.291)
Shock $\times$ Q3			1.223 (0.180)			1.903 (0.800)
Shock $\times$ Q4			1.906*** (0.409)			3.286** (1.904)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Failures	18,433	18,433	18,433	3,252	3,252	3,252
Displacement cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Shock $\times$ RI^w	No	Yes	Quartile	No	Yes	Quartile

Notes: Cause-specific Cox proportional hazard ratios reported throughout. $HR > 1$ indicates faster reemployment relative to the baseline group. Immediate reemployment is defined as reemployment in the same calendar month as displacement (spell length = 1 month); non-immediate reemployment requires at least one full calendar month of non-employment. Workers not reemployed by December 2012 are right-censored. Displacement cohort fixed effects (year of last construction employment, 2007–2012) are included in all specifications to absorb mechanical differences in the observation window: high-index workers exit construction later on average, shortening their window before the censoring date. Columns (1) and (4) include RI^w as a main effect only. Columns (2) and (5) add the shock main effect and the Shock $\times$ RI^w interaction. Columns (3) and (6) replace the continuous interaction with shock interactions for each quartile of the RI^w distribution. Efron method for tied failures. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C10: Reallocation Index and Labor Market Outcomes, Leave-One-Province-Out Index

	(1)	(2)	(3)
	Cumulative earnings	Cumulative employment	Avg. wage (cond.)
Shock	−2.040*** (0.278)	−1.613*** (0.155)	−0.013* (0.007)
Reallocation index	−0.089 (0.094)	−0.143*** (0.049)	0.000 (0.002)
Shock × Index	0.339** (0.158)	0.334*** (0.078)	0.002 (0.003)
Baseline controls	Yes	Yes	Yes
Observations	34,000	34,000	34,000

Notes: Dependent variables are normalized cumulative earnings (Column 1), cumulative days employed annualized (Column 2), and average monthly earnings conditional on employment (Column 3), each measured over 2007–2012 and normalized by the worker’s average annual earnings in 2005–2006. The reallocation index is constructed using a leave-one-province-out procedure: for workers in province p , the multinomial logit transition probabilities are estimated on workers from all other provinces and applied to workers in p , ensuring the index cannot encode province-specific selection patterns. The leave-out index has a correlation of 0.964 with the baseline index. Provincial employment shares are unchanged. Standard errors are clustered at the province level.

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Online Appendix

OA.A Data Appendix

What follows complements the exposition in the main text. Section OA.A.1 describes the construction of the estimation sample and documents the bite of each restriction. Section OA.A.2 defines the individual, contract, and regional controls that enter the baseline specification. Section OA.A.3 details the construction of the reallocation index RI_i^w , covering the multinomial logit training sample, the sector probability predictions, the location-quotient normalization, and the wage-ratio component. Section OA.A.4 collects the formal derivations underlying the index.

OA.A.1 Sample Construction

The estimation sample is drawn from the MCVL and applies four restrictions sequentially to the available sample of workers observed in Social Security records between 2006 and 2021.³²

The complete panel covers all workers affiliated with Social Security between the ages of 18 and 60 across all 50 Spanish provinces, excluding the autonomous cities of Ceuta and Melilla. Province identifiers coded as zero or missing are dropped throughout; workers whose province of residence cannot be determined are excluded from both the estimation sample and the mlogit training sample described in Section OA.A.3. The panel is collapsed to a monthly frequency: for workers with multiple simultaneous spells in a given month, the record with the highest earnings is retained.

The first restriction defines sectoral attachment. A worker is included in the estimation sample if she accumulates at least six calendar months of employment in the construction sector during 2005 and 2006 combined. A month counts toward this threshold if the worker’s primary employment spell in that month is a construction spell. The six-month threshold targets workers with established sectoral attachment—the sample-mean construction tenure in 2007 is 11 months—and excludes workers who entered construction only during the boom’s final expansion.

The second restriction requires that the worker be observed employed in construction in at least one month of 2007. This ensures that all sample members were active in the sector at the onset of the shock and excludes workers who had already left construction before the

³²The information in the MCVL is a 4% random sample of workers affiliated with the Social Security in the year of reference and all their previous labor market history. I use the MCVL editions from 2006–2021, which include all individuals in 2006 plus all individuals who appear in subsequent years as replacements for individuals leaving the Social Security records.

crisis began. The resulting crosswalk of qualifying `person_id` values is saved separately and used as an inner-join key for all subsequent merges.

The third restriction limits the sample to native-born workers. Foreign-born workers constituted 27 percent of construction employment in 2007 and exhibit higher rates of return migration during downturns. Because return migration is concentrated in high-exposure provinces, differential attrition would contaminate the cross-province comparison that identifies the shock effect. Workers with a foreign-born indicator equal to one are dropped before the outcome variables are constructed.

The fourth restriction excludes workers aged above 50 in the baseline year 2007, isolating labor-market adjustment from retirement transitions.³³ Age is measured as of January 2007, and the restriction is applied after collapsing the panel to one observation per worker in 2007.

After applying all four restrictions, the panel is collapsed to a worker-level cross-section at the 2007 baseline. For each worker the first employment record in 2007 is retained; all individual and job characteristics are measured from this record. Province of residence is assigned the province of the primary 2007 employer, which serves as the fixed location for computing the shock variable and the reallocation index. The resulting cross-section contains exactly 34,000 workers.

Cumulative outcomes are constructed from the full 2007–2012 panel before collapsing. Monthly earnings from tax records are converted to a monthly equivalent of annual earnings; months of non-employment contribute zero to both the earnings and the days-worked sums. Cumulative days worked and cumulative earnings are then normalized by the worker’s average annual earnings during 2005–2006 to obtain outcomes that are comparable across workers with different pre-crisis earnings levels. For workers whose 2005–2006 average earnings are missing—a small share arising from late labor-market entry—the normalizing denominator is replaced by the first observed monthly earnings, converted to an annual equivalent.

OA.A.2 Control Variables

The baseline specification in equation (2) of the main text includes three groups of predetermined controls. The first group captures life-cycle heterogeneity. Age is partitioned into ten four-year intervals from 18 to 57 plus a top-coded category for workers aged 57 and above. These age indicators are interacted fully with gender and educational attainment—below secondary, secondary, or tertiary. The interactions absorb all main effects of gender and

³³The upper bound is set at 50 rather than 55 because construction workers in their early 50s exhibit elevated rates of transition into disability insurance in the post-crisis period—a margin of adjustment that is not the object of study here. Results are robust to extending the ceiling to age 55.

education, and the same age-category grid is used in the mlogit training sample to ensure that predicted sector probabilities $\hat{\phi}_j(X_i)$ are applied out-of-sample on an identical covariate space.

The second group controls for job and contract characteristics measured at the 2007 baseline. Occupational skill group is included as a set of five indicators corresponding to the Social Security classification, ordered from very-high-skilled to low-skilled. Tenure, measured in years from the start of the current employment spell to January 2007, enters linearly and proxies for job stability at the time of the shock; workers with missing tenure are excluded from the estimation sample before any other restriction is applied. Labor-market experience is computed as the cumulative number of months in which the worker appears in Social Security records since labor-market entry up to 2007, expressed in years, and enters the specification together with its square. Two contract-type indicators—one for fixed-term contracts and one for part-time employment, both drawn from the primary 2007 spell—capture differences in pre-crisis employment conditions that may predict adjustment independently of occupation and tenure. Finally, earnings quintiles are constructed from the distribution of monthly earnings in 2007 within the estimation sample and enter as a set of four dummies, absorbing residual variation in pre-crisis earnings levels not explained by the other job characteristics.

The third group consists of province-level controls, all measured in 2006 and merged to the cross-section on the province identifier. The initial construction employment share $s_{r,2006}^{cs}$ ensures that the shock coefficient does not capture variation mechanically related to the sector’s pre-crisis size. The 2006 provincial unemployment rate, drawn from registered unemployment records, absorbs pre-existing differences in labor-market tightness. A Bartik demand shifter aggregates national employment growth rates in non-construction sectors, weighted by each sector’s 2006 employment share in province r , and controls for demand trends in surviving sectors that could affect reemployment outcomes independently of the construction collapse. The Herfindahl–Hirschman index of non-construction sectoral concentration, $HHI_r = \sum_{j \neq cs} (s_{jr})^2$, absorbs variation in local economic diversity. Provinces with highly concentrated non-construction employment may offer fewer alternative destinations for all workers, regardless of individual skill profiles.

OA.A.3 Construction of the Reallocation Index

The reallocation index RI_i^w summarizes the expected arrival rate of acceptable job offers from surviving non-construction sectors for worker i in province r . Formally,

$$RI^w(X_i, r) = \sum_{j \neq cs} \hat{\phi}_j(X_i) \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^{2\eta} \cdot \frac{w_j}{w_{cs}}, \quad (A1)$$

where $\hat{\phi}_j(X_i)$ is the predicted probability that a worker with characteristics X_i is employed in sector j ; $(s_{jr}/\bar{s}_j)^{2\eta}$ is the location quotient of sector j in province r raised to a power governed by the matching elasticity η ; and w_j/w_{cs} is the ratio of sector j 's national median wage to the construction median, both measured in 2006. The exponent 2η is derived in Section OA.A.4; the baseline sets $\eta = 1/2$, so the location quotient enters linearly. Without loss of generality, any $\eta \in (0, 1)$ preserves the index's monotonicity in local specialization and alters only the curvature with which specialization enters, not its direction.³⁴

The MCVL sectoral classification comprises thirteen sectors; construction is sector 5. Construction is excluded at every step of the index, so all summations and the multinomial logit below run over the remaining twelve non-construction sectors. Each component is constructed separately and then combined multiplicatively for each worker–sector–province triple; construction contributes zero by design. The index is normalized to mean zero and unit standard deviation in the estimation sample before entering any regression. The three-step construction proceeds as follows.

Step 1: Sector Transition Probabilities The predicted probabilities $\hat{\phi}_j(X_i)$ are estimated from a multinomial logit model fit to pre-recession employment transitions by non-construction workers. The training sample covers the period 2002–2006 and applies three restrictions. First, it retains only workers employed outside the construction sector, so that the estimated coefficients reflect sector-entry probabilities. Second, public-sector workers are dropped; only private-sector employment spells are retained. Third, only the *first observation at each new employer* is kept: the sample is restricted to records for which the establishment identifier differs from the preceding record of the same worker, and tenure at that establishment is less than one month. This restriction targets the moment of sector entry rather than sectoral incumbency. A worker who has been employed in manufacturing for three years reveals little about the characteristics associated with just entering manufacturing. By conditioning on job-entry spells, the mlogit coefficients reflect the relation-

³⁴The implementation in equation (A23) uses the linear baseline $2\eta = 1$. Reporting the general form here makes explicit that the linear case is a normalization of the matching elasticity rather than a separate functional-form assumption.

ship between observable characteristics and the probability of landing in each sector when changing employers, which is the relevant margin for displaced workers searching for new employment.

The covariate vector X_i in the mlogit includes gender interacted with the ten age-category indicators, a foreign-born dummy, age categories interacted with education, and occupational skill group. The age-category binning in the training sample matches exactly the binning defined above, ensuring that the estimated coefficients can be applied out-of-sample to the estimation sample on an identical covariate grid. The base outcome is sector 3 (manufacturing of non-durable goods). Province fixed effects are not included. The transition probabilities are intended to capture skill-based sector fit common across provinces, not province-specific demand conditions that are separately controlled for by the local specialization component.

The estimated mlogit coefficients are stored and then applied to the 2007 cross-section of construction workers. For each worker i and each non-construction sector j , the predicted probability $\hat{\phi}_j(X_i)$ is the multinomial logit choice probability evaluated at worker i 's observed characteristics. Construction (sector 5) is excluded from the prediction set, so $\hat{\phi}_5(X_i) = 0$ for all i by construction.

Because the mlogit is estimated on pre-recession data (2002–2006) and applied to the post-shock period, a concern arises that skill–sector mappings may have shifted during the downturn. Re-estimating the mlogit on post-recession job-entry spells (2010–2015) yields a reallocation index with a rank correlation of $\rho = 0.873$ relative to the baseline, indicating that the structural relationship between observable characteristics and sector entry is largely stable across the two periods.

Step 2: Sectoral Specialization Ratios The local opportunity component $s_{jr}/\bar{s}_j$ measures the relative concentration of sector j in province r . The province-level employment shares s_{jr} are computed from the MCVL using a single cross-section of March 2006, a choice motivated by two considerations. First, using a single month rather than an annual average avoids seasonality in employment compositions that would otherwise contaminate the location quotients; March is sufficiently distant from the holiday seasons that affect hospitality, agriculture, and retail sectors. Second, measuring shares in 2006 ensures that the opportunity set is predetermined with respect to the post-2007 shock. Construction workers are excluded from both the numerator and the denominator when computing s_{jr} , so that the shares reflect the sectoral composition of the non-construction labor market that displaced workers will search in.

Formally, for each sector $j \neq \text{cs}$ and province r , let E_{jr} denote the number of non-construction workers employed in sector j in province r in March 2006, and let $E_r =$

$\sum_{k \neq cs} E_{kr}$ be total non-construction employment in province r . Then

$$s_{jr} = \frac{E_{jr}}{E_r}. \quad (\text{A2})$$

The national share $\bar{s}_j$ is computed over the same population and under the same restrictions used to construct s_{jr} , so that the location quotient $s_{jr}/\bar{s}_j$ compares like with like. The resulting shares $\bar{s}_j$ represent the national composition of non-construction employment in the pre-recession period, which is the appropriate benchmark for assessing whether a given province is over- or under-represented in each sector relative to the employment structure that displaced workers can realistically access.

The specialization ratio $s_{jr}/\bar{s}_j$ takes values above one for sectors that are over-represented in province r relative to the national average, and below one for under-represented sectors. Dividing by $\bar{s}_j$ prevents large sectors from mechanically dominating the index by virtue of their national scale. Construction's location quotient is set to zero for all provinces, reflecting the assumption that the sector no longer constitutes a viable reemployment destination after the permanent demand contraction.

Step 3: Sector Wage Medians The wage component w_j/w_{cs} captures the probability that a displaced construction worker accepts an offer from sector j , proxying for the acceptance margin in the search framework developed in Section [OA.A.4](#). Higher-paying sectors are assumed to generate offers more likely to clear the worker's reservation wage, which is anchored to pre-displacement construction earnings.

Sector wage medians w_j are computed from the 2006 cross-section of the estimation sample, using the primary monthly earnings record for each worker in their observed sector. Measuring medians in 2006 ensures the wage component is predetermined with respect to the post-2007 shock. Medians are computed at the national level rather than the province level for two reasons. First, national medians capture sector-specific skill requirements common across regions rather than local cost-of-living differences unrelated to job quality. Second, province-level medians would be estimated on small cells for several sector-province pairs, introducing noise that would propagate into the index. Each sector median is computed as the median of individual earnings across all workers observed in that sector in 2006, regardless of whether they are in the estimation sample. The construction median w_{cs} is computed analogously from construction workers in the same year. All medians are normalized by the construction median, so $w_{cs}/w_{cs} = 1$ by construction.

OA.A.4 Additional Theoretical Results

Derivation of Restriction A1. The Bellman equation for a worker employed in sector j at wage $w_{jr}(X_i)$ is

$$\rho W_j(X_i) = w_{jr}(X_i) + \delta [U(X_i, r) - W_j(X_i)], \quad (\text{A3})$$

where jobs are destroyed at rate δ and the worker returns to unemployment with value $U(X_i, r)$. Solving for $W_j(X_i)$:

$$W_j(X_i) = \frac{w_{jr}(X_i) + \delta U(X_i, r)}{\rho + \delta}. \quad (\text{A4})$$

Defining $S_j(X_i) \equiv W_j(X_i) - U(X_i, r)$ and substituting in (A4):

$$S_j(X_i) = \frac{w_{jr}(X_i) - \rho U(X_i, r)}{\rho + \delta}. \quad (\text{A5})$$

The Nash bargaining wage and productivity function give $w_{jr}(X_i) = \beta a_j \phi_j(X_i) + (1 - \beta)b$, so

$$S_j(X_i) = \frac{\beta a_j \phi_j(X_i) + (1 - \beta)b - \rho U(X_i, r)}{\rho + \delta}. \quad (\text{A6})$$

Equation (A6) is not yet linear in $\phi_j(X_i)$ because $U(X_i, r)$ depends on the full distribution of surpluses through

$$\rho U(X_i, r) = b + \max_{\{\sigma_j\}} \sum_{j \neq cs} \sigma_j(X_i, r) \cdot p_{jr} \cdot S_j(X_i). \quad (\text{A7})$$

The surplus S_j thus appears on both sides of the system, precluding a closed-form solution without further restriction.

Let $\bar{\phi}_j$ denote average match quality and $\bar{U}(r)$ the unemployment value evaluated at the average. A first-order Taylor expansion of (A6) around $\phi_j(X_i) = \bar{\phi}_j$ gives

$$S_j(X_i) \approx \underbrace{\frac{\beta a_j \bar{\phi}_j + (1 - \beta)b - \rho \bar{U}(r)}{\rho + \delta}}_{\bar{S}_j(r)} + \frac{\beta a_j}{\rho + \delta} [\phi_j(X_i) - \bar{\phi}_j]. \quad (\text{A8})$$

The second-order correction is proportional to $\beta a_j/(\rho + \delta)$ times an equilibrium search-allocation weight and is therefore negligible for small deviations $\phi_j(X_i) - \bar{\phi}_j$. The intercept $\bar{S}_j(r) - \beta a_j \bar{\phi}_j/(\rho + \delta)$ varies across sectors but is constant across workers within a sector, depending only on average match quality $\bar{\phi}_j$ and the province-level unemployment value $\bar{U}(r)$; it shifts all workers' valuations of sector j equally and drops out when ranking sectors

by surplus. The surplus is therefore proportional to match quality:

$$S_j(X_i) \approx \frac{\beta}{\rho + \delta} \cdot a_j \cdot \phi_j(X_i). \quad (\text{A9})$$

Derivation of Restriction A2. The free-entry condition equates the flow cost of a vacancy to the expected return from filling it. From Nash bargaining the firm captures fraction $(1 - \beta)$ of the average match surplus $\bar{S}_j$, so the firm value of a filled position is $J_j = (1 - \beta) \bar{S}_j$. Substituting the vacancy-filling rate $q_{jr} = m_0 \theta_{jr}^{\eta-1}$ into the free-entry condition and solving for tightness:

$$\theta_{jr}^{\text{nat}} = \left(\frac{m_0 (1 - \beta) \bar{S}_j}{\kappa_j} \right)^{1/(1-\eta)}. \quad (\text{A10})$$

This expression is province-invariant, capturing only the national-level attractiveness of sector j and implying uniform vacancy posting across provinces. Vacancy posting is instead more attractive where sector j is locally concentrated, since a thicker pool of suitable workers raises the expected return to posting. Let s_{jr} denote the employment share of sector j in province r and $\bar{s}_j$ its national counterpart, so that $s_{jr}/\bar{s}_j > 1$ identifies provinces specialized in sector j .³⁵ Incorporating local specialization into the free-entry condition yields equilibrium tightness

$$\theta_{jr}^* = \left(\frac{m_0 (1 - \beta) \bar{S}_j}{\kappa_j} \right)^{1/(1-\eta)} \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right). \quad (\text{A11})$$

Substituting (A11) into the offer-receipt probability $p_{jr} = m_0 \theta_{jr}^\eta$:

$$p_{jr} = m_0 \underbrace{\left(\frac{m_0 (1 - \beta) \bar{S}_j}{\kappa_j} \right)^{\eta/(1-\eta)}}_{A_j} \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^\eta, \quad (\text{A12})$$

where A_j is a sector-specific constant absorbing all province-invariant parameters. The offer probability is proportional to $(s_{jr}/\bar{s}_j)^\eta$, so sectors with greater local specialization generate faster offer arrivals.

Derivation of Restriction A3. Worker i accepts a sector- j offer if $W_j(X_i) \geq U(X_i, r)$, equivalently if $S_j(X_i) \geq 0$. With reservation wages $w_i^R = \mu + \varepsilon_i$ and $\varepsilon_i \sim F$, the aggregate acceptance probability for a sector- j offer is

$$\alpha_j(X_i) = F(w_j - \mu). \quad (\text{A13})$$

³⁵The assumption requires only that $s_{jr}/\bar{s}_j$ enter the free-entry condition monotonically; any functional form preserving this property yields the same equilibrium tightness expression.

Under a uniform distribution of ε_i on $[\underline{\varepsilon}, \bar{\varepsilon}]$ with support spanning the sectoral wage distribution, $\alpha_j(X_i)$ is linear in w_j .³⁶ Normalizing by the pre-displacement wage w_{cs} gives

$$\alpha_j(X_i) \propto \frac{w_j}{w_{cs}}, \quad (\text{A14})$$

so sectors paying closer to pre-displacement earnings attract higher acceptance rates, capturing the wage-adequacy margin of the reemployment problem.

OA.A.4.1 Derivation of the Reallocation Index

The flow rate of acceptable offers from sector j for worker i is the product of three terms: search intensity directed at sector j , the offer arrival rate conditional on search, and the probability that a received offer is acceptable:

$$\lambda_{ij}(r) = \sigma_j^*(X_i, r) \cdot p_{jr} \cdot \alpha_j(X_i). \quad (\text{A15})$$

Summing across sectors gives the total arrival rate of acceptable offers:

$$\Lambda_i(r) = \sum_{j \neq cs} \sigma_j^*(X_i, r) \cdot p_{jr} \cdot \alpha_j(X_i). \quad (\text{A16})$$

Under Restriction A3, the acceptance probability is proportional to the wage ratio, $\alpha_j(X_i) \propto w_j/w_{cs}$. Substituting into (A16):

$$\Lambda_i(r) = \sum_{j \neq cs} \sigma_j^*(X_i, r) \cdot p_{jr} \cdot \frac{w_j}{w_{cs}}. \quad (\text{A17})$$

Under Restriction A2, equilibrium tightness satisfies $\theta_{jr}^* = C_j \cdot (s_{jr}/\bar{s}_j)$, where $C_j = (m_0 \bar{S}_j / \kappa_j)^{1/(1-\eta)}$ collects national-level sector constants. The offer arrival rate is $p_{jr} = m_0 \cdot \theta_{jr}^\eta$, so

$$p_{jr} = m_0 \cdot C_j^\eta \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^\eta \equiv A_j \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^\eta, \quad (\text{A18})$$

where $A_j \equiv m_0 \cdot C_j^\eta$ is province-invariant. Substituting into (A17):

$$\Lambda_i(r) = \sum_{j \neq cs} \sigma_j^*(X_i, r) \cdot A_j \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^\eta \cdot \frac{w_j}{w_{cs}}. \quad (\text{A19})$$

Under Restriction A1, $S_j(X_i) \approx \frac{\beta}{\rho+\delta} a_j \phi_j(X_i)$. The optimal search intensity is

³⁶Monotonicity of α_j in w_j holds for any continuous F . The uniform specializes this to linearity, yielding the proportionality in equation (A14) without additional free parameters.

$\sigma_j^*(X_i, r) = p_{jr} \cdot S_j(X_i)$, so substituting A1 and A2:

$$\sigma_j^*(X_i, r) = A_j \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^\eta \cdot \frac{\beta}{\rho + \delta} \cdot a_j \cdot \phi_j(X_i) \equiv \tilde{B}_j \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^\eta \cdot \phi_j(X_i), \quad (\text{A20})$$

where $\tilde{B}_j \equiv A_j \cdot a_j \cdot \beta / (\rho + \delta)$ is province-invariant. Substituting (A20) into (A19):

$$\Lambda_i(r) = \sum_{j \neq \text{cs}} \tilde{B}_j \cdot A_j \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^{2\eta} \cdot \phi_j(X_i) \cdot \frac{w_j}{w_{\text{cs}}}. \quad (\text{A21})$$

The specialization ratio enters with exponent 2η . Setting $\eta = 1/2$ delivers $2\eta = 1$, so the ratio enters linearly. Any $\eta \in (0, 1)$ preserves monotonicity of the index in local specialization, so this restriction affects the curvature of the relationship but not its direction. [Petrongolo and Pissarides \(2001\)](#) find that estimates of η cluster between 0.5 and 0.7 across a wide range of countries and time periods, making $\eta = 1/2$ an empirically grounded baseline. The sector-specific constants $\tilde{B}_j \cdot A_j$, which are province-invariant, are absorbed into the alternative-specific intercepts of the multinomial logit used to estimate $\phi_j(X_i)$ and play the role of Λ_0 in the empirical implementation. The arrival rate therefore reduces to

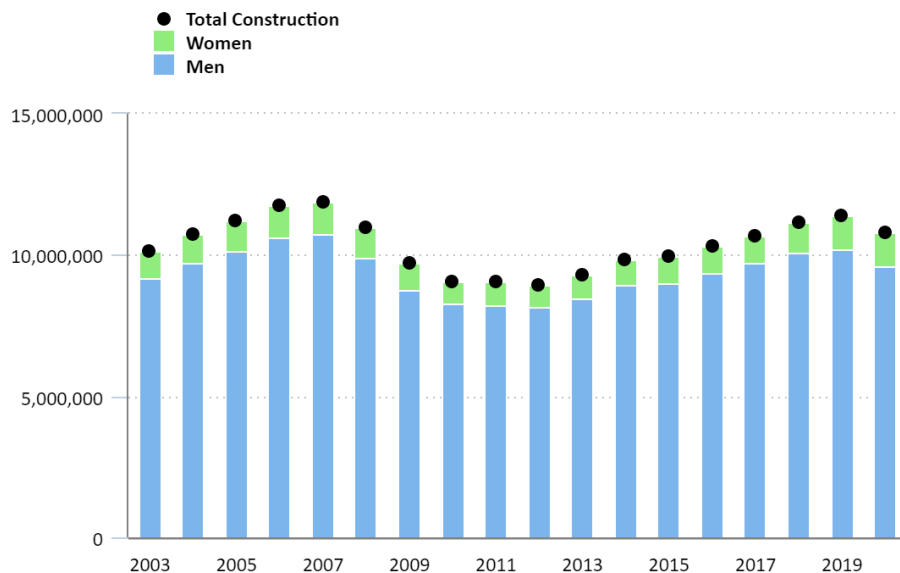
$$\Lambda_i(r) = \Lambda_0 \cdot \sum_{j \neq \text{cs}} \phi_j(X_i) \cdot \frac{s_{jr}}{\bar{s}_j} \cdot \frac{w_j}{w_{\text{cs}}}, \quad (\text{A22})$$

and the reallocation index is defined as the model-implied arrival rate normalized by Λ_0 :

$$\text{RI}^w(X_i, r) \equiv \frac{\Lambda_i(r)}{\Lambda_0} = \sum_{j \neq \text{cs}} \phi_j(X_i) \cdot \frac{s_{jr}}{\bar{s}_j} \cdot \frac{w_j}{w_{\text{cs}}}. \quad (\text{A23})$$

OA.B Additional Figures

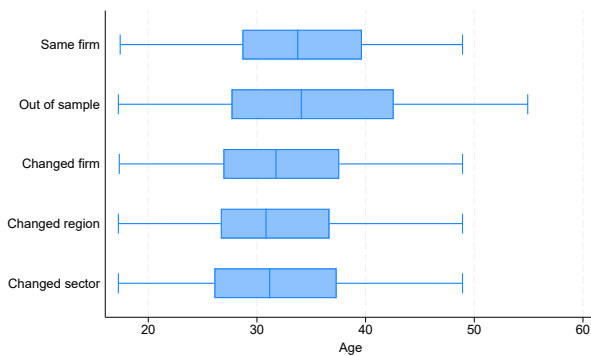
Figure OA.1: Construction sector employment in the US, 2003–2020



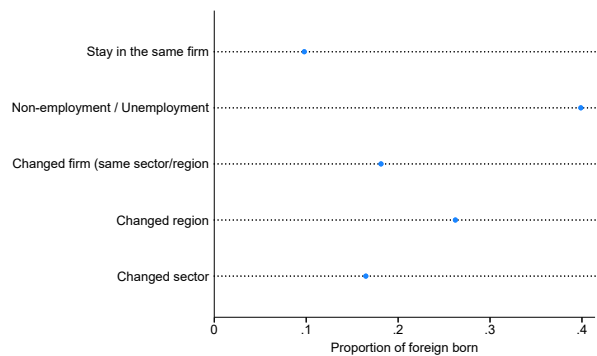
Source: U.S. Bureau of Labor Statistics.

Notes: Construction employment by gender in the US, 2003–2020. *Source:* US Bureau of Labor Statistics.

Figure OA.2: Characteristics of workers initially in the construction sector by employment status in 2013



(a) Average age



(b) Share of foreign workers

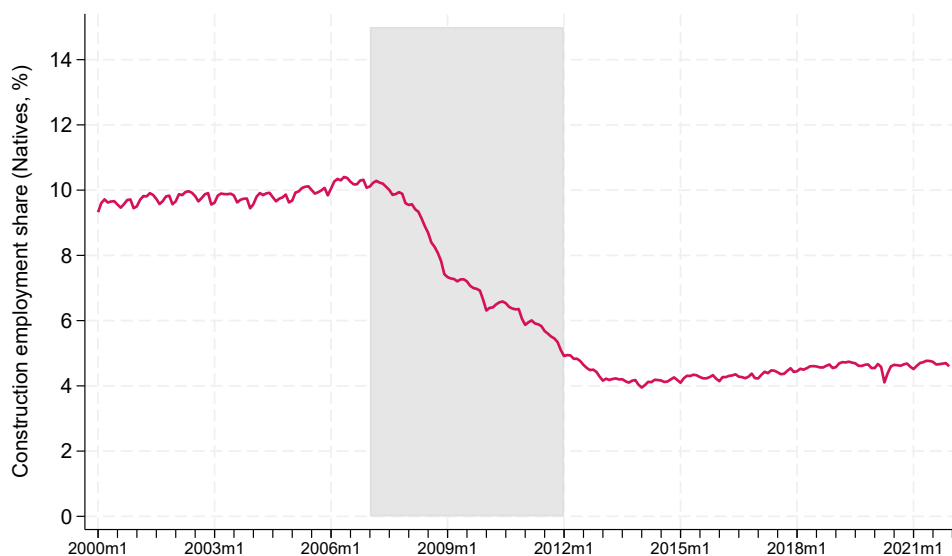
Notes: Panel (a) average age in 2007 of workers in the construction sector by status in 2013. Panel (b) share of foreign workers in the construction sector by status in 2013. The sample is restricted to workers in the construction sector in 2007 aged 20–55 years old.

Figure OA.3: Employment share of male workers in the construction sector, 1995–2019



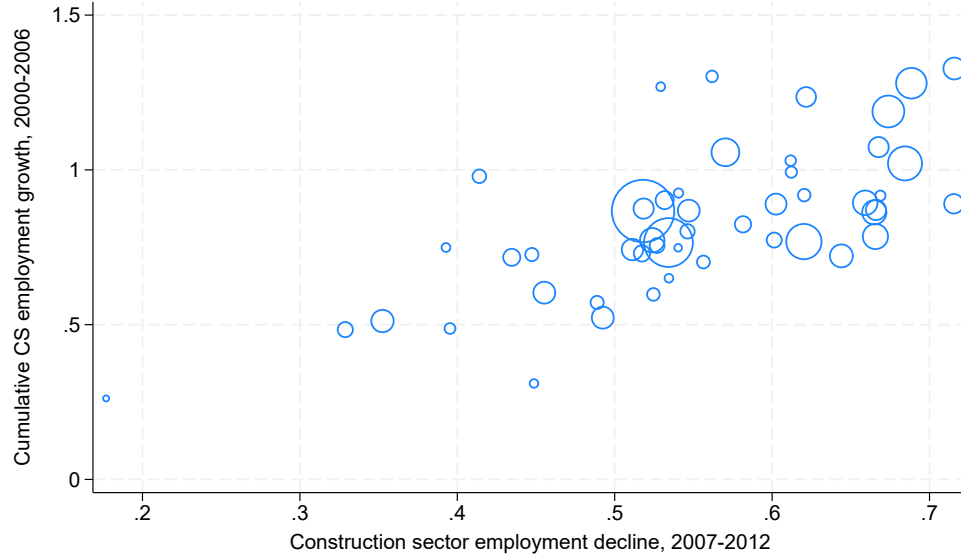
Notes: Presents the proportion of male workers in Spain's construction sector from January 2000 to December 2017. The data are restricted to male workers aged 20–60 and employed during the referenced period. The shaded area comprises the years of the Great Recession in Spain, between 2007 and 2012.

Figure OA.4: Employment share of native workers in the construction sector, 1995–2019



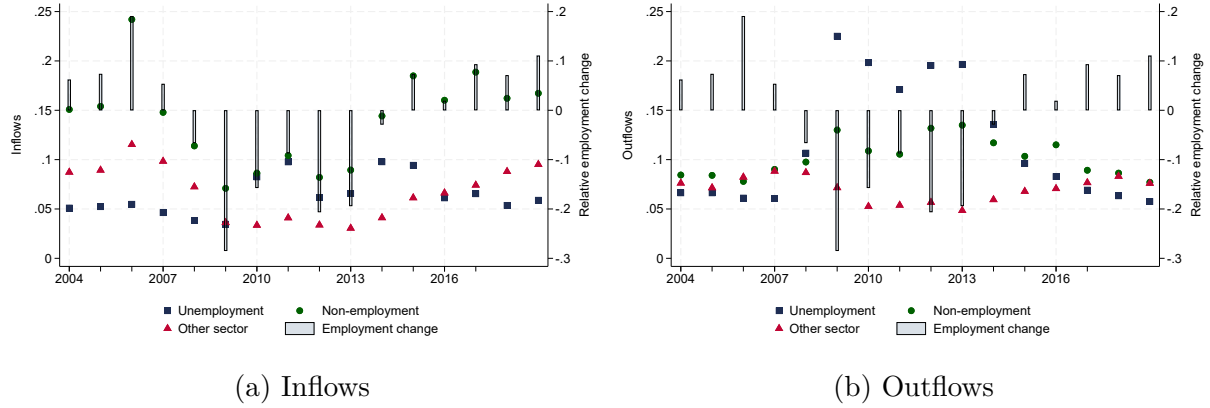
Notes: Presents the proportion of native workers in Spain's construction sector from January 2000 to December 2017. The data are restricted to native workers aged 20–60 and employed during the referenced period. The shaded area comprises the years of the Great Recession in Spain, between 2007 and 2012.

Figure OA.5: Cumulative growth in the construction sector, 2000–2006



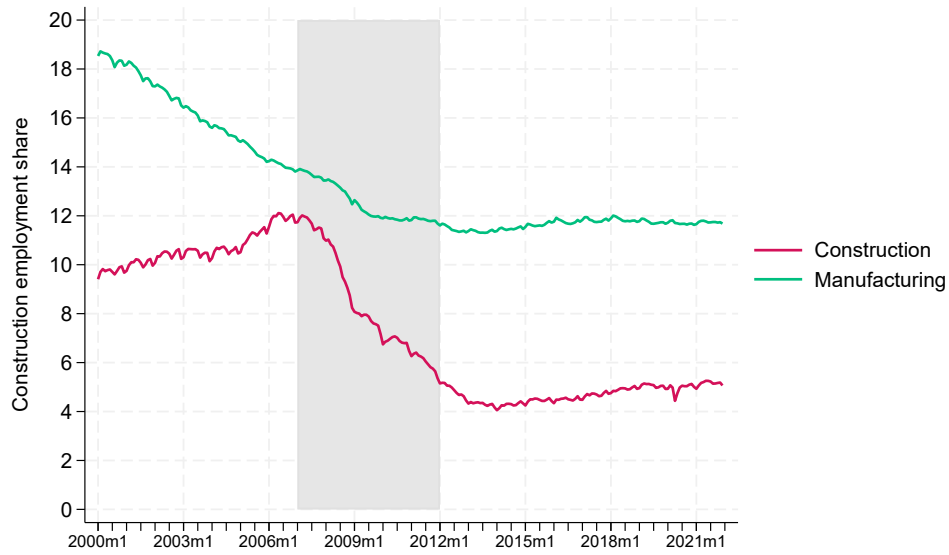
Notes: Change in the employment share of the construction sector by province between 2007 and 2012 against cumulative employment share growth in the construction sector between 2000 and 2006, weighted by employment in the construction sector in 2006.

Figure OA.6: Aggregate flows from and to the construction sector



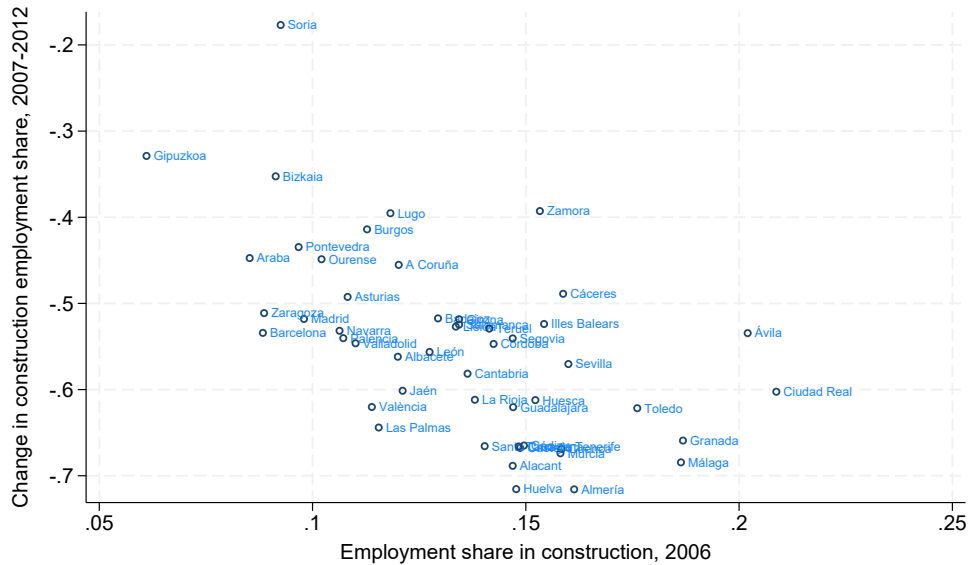
Notes: Panel (a) reports inflows into the construction sector, defined as the share of individuals who transition into construction from another sector, non-employment, or unemployment between years $t - 1$ and t , relative to total employment in year $t - 1$. Panel (b) reports outflows from the construction sector, defined as the share of workers employed in construction in year $t - 1$ who transition to another sector, non-employment, or unemployment in year t . The sample consists of annual observations from 2003 to 2019 for workers aged 18–60.

Figure OA.7: Manufacturing and construction employment shares, 1995–2019



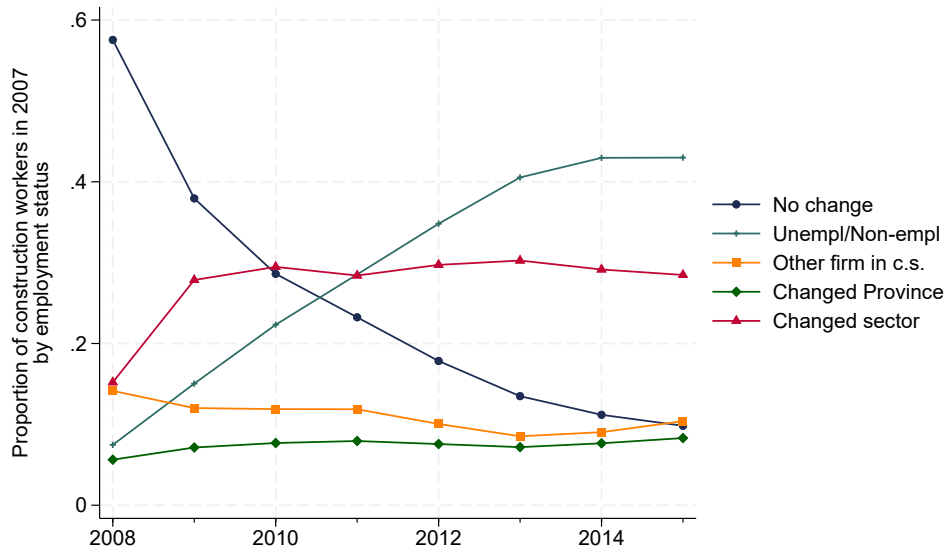
Notes: Employed workers in the construction and manufacturing sectors as a percentage of all employed workers, January 1995 to December 2019. Data are restricted to workers employed during the referenced period. The shaded area comprises the years of the Great Recession in Spain, between 2008 and 2014.

Figure OA.8: Change in the construction employment share by province, 2003–2020



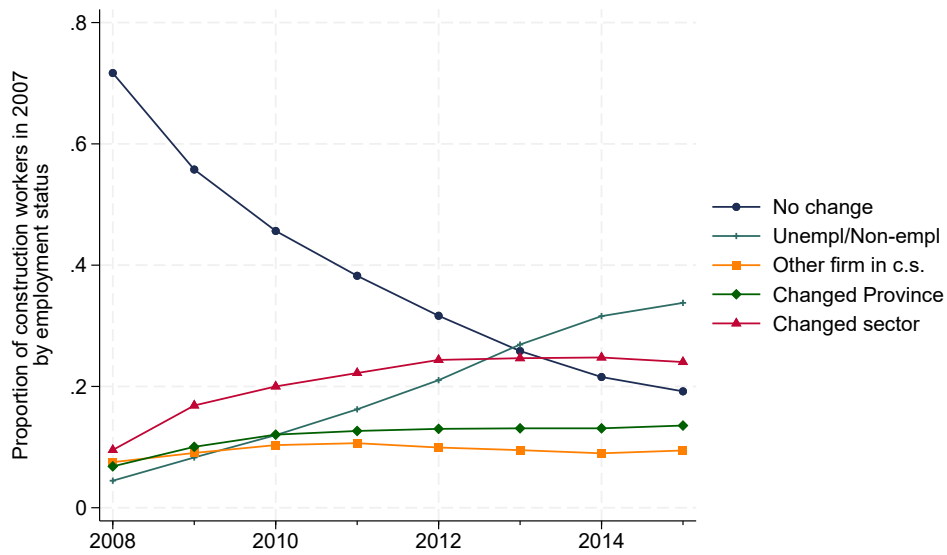
Notes: Proportional change in the employment share of the construction sector by province between 2007 and 2012 against construction employment share in 2006. The sample considers the 50 Spanish provinces.

Figure OA.9: Working status of individuals employed in the construction sector in 2007



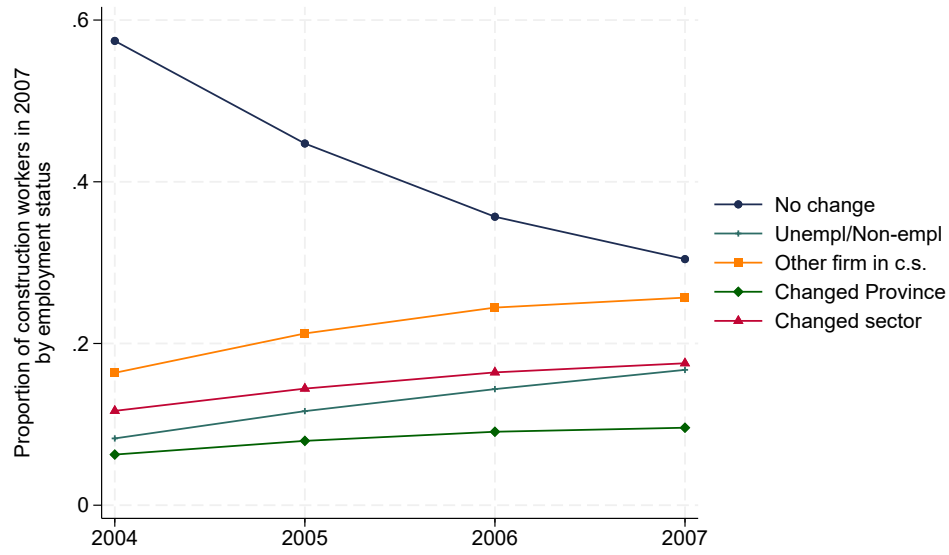
Notes: The sample includes native workers employed in the construction sector in 2007. Employment status is tracked annually through 2015.

Figure OA.10: Working status of highly skilled individuals in 2007



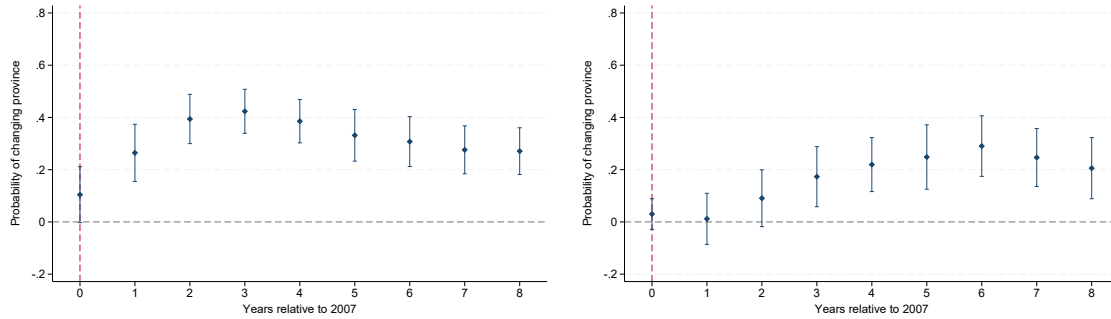
Notes: The shares are computed based on highly skilled workers; every year, I track their working status up to 2015.

Figure OA.11: Working status of workers in the construction sector in 2003



Notes: The shares are computed based on native workers employed in the construction sector in 2003; every year I track their working status up to 2007.

Figure OA.12: Probability of Sectoral versus Geographic Reallocation

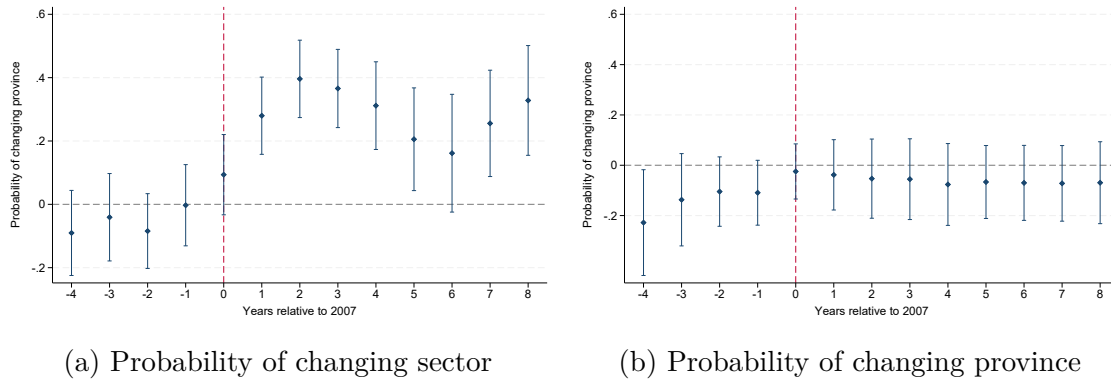


(a) Probability of changing sector

(b) Probability of changing province

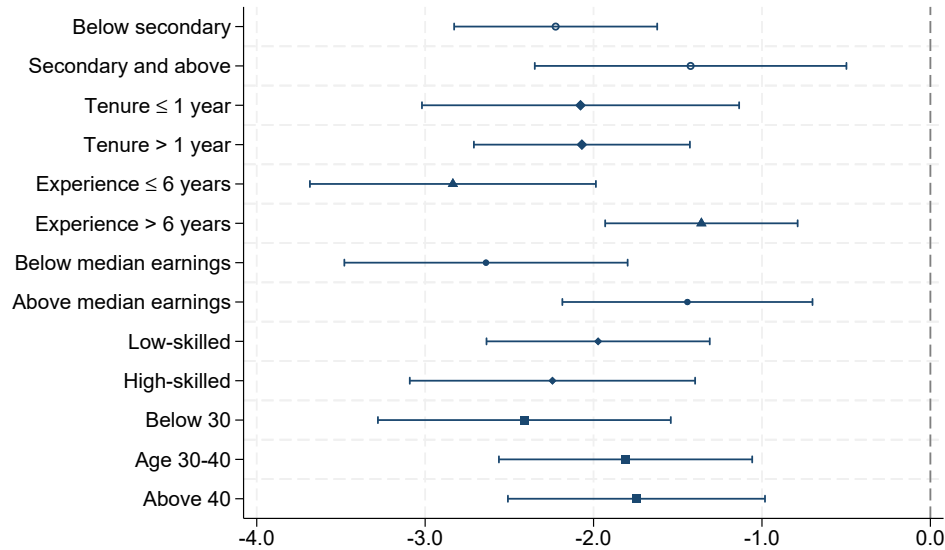
Notes: Sample restricted to workers aged 29–42 employed in construction in 2007. Panel (a): probability of transitioning to a different sector; Panel (b): probability of residing in a different province from the 2007 province of residence. Additional controls include the initial construction employment share, a Bartik-type demand shifter, and demographic characteristics. Outcomes are cumulative changes in province or sector.

Figure OA.13: Probability of Sectoral versus Geographic Reallocation (Robustness)



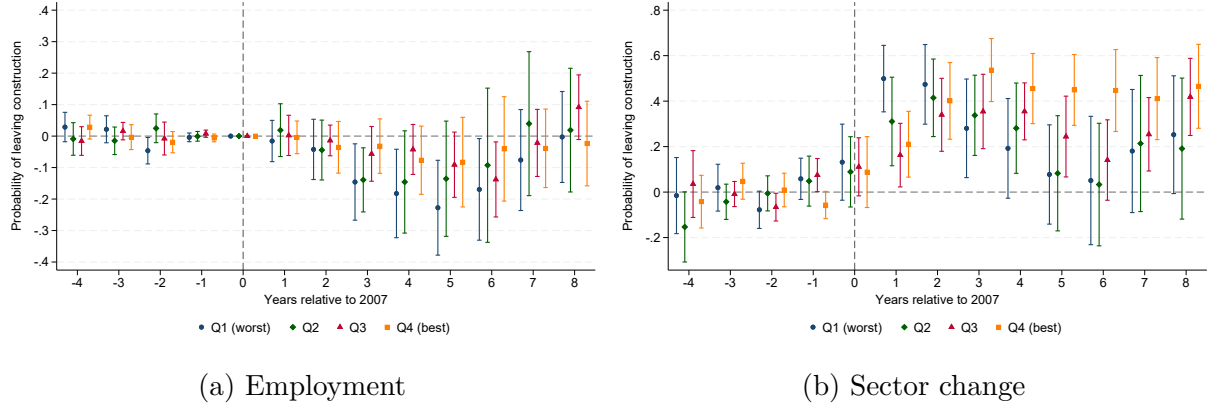
Notes: Panel (a): probability of transitioning to a different sector; Panel (b): probability of residing in a different province from the 2007 province of residence.

Figure OA.14: Heterogeneity of the Shock's Impact on Earnings Losses by Worker Characteristics



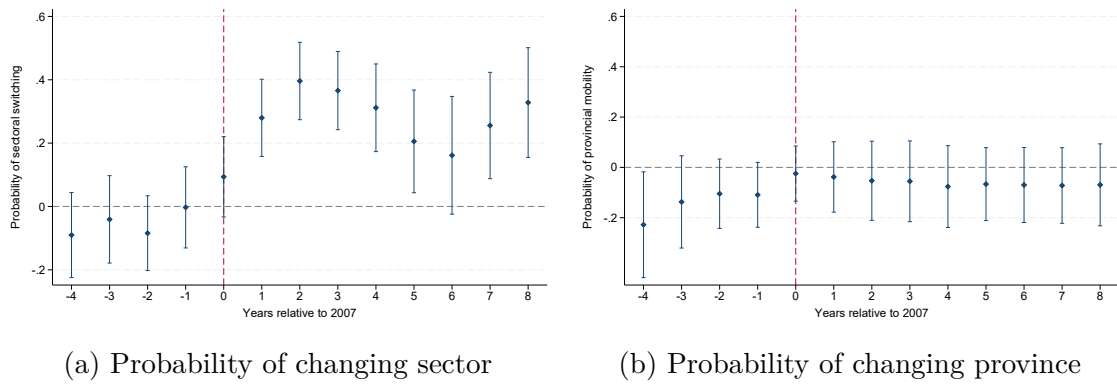
Notes: Estimated earnings losses by baseline worker and employment characteristics. Sample restrictions and control variables are identical to those used in Table 1. Standard errors are clustered at the province level.

Figure OA.15: Probability of Employment and Sectoral Exit by Reallocation Index Quartile



Notes: Year-by-year linear probability estimates. Dependent variable: indicator for employment in a sector other than construction in year t , for $t \in \{2003, \dots, 2015\}$. Each year is estimated separately as a regression of the annual sector-change indicator on the shock interacted with indicators for each quartile of the reallocation index, conditional on quartile fixed effects and the pre-shock (2003–2006) average of the sector-change indicator. Controls and sample restrictions follow Table 1. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure OA.16: Probability of Sectoral versus Geographic Reallocation (Unconditional)



Notes: Panel (a) reports the probability of transitioning out of the construction sector into a different sector; Panel (b) reports the probability of residing in a province different from the 2007 province of residence. The sample and control variables follow the baseline specification in Table 1.

OA.C Additional Tables

Table OA.1: Characteristics of Construction Sector Workers, 2004–2017

	Year			
	2004	2007	2012	2017
<i>Age distribution</i>				
Average age	34.02	34.73	38.24	40.93
<i>Education</i>				
Below secondary	0.758	0.746	0.656	0.671
Secondary	0.156	0.161	0.199	0.189
Tertiary	0.086	0.092	0.146	0.141
<i>Contract characteristics</i>				
Part-time	0.037	0.035	0.074	0.088
Fixed-term	0.724	0.659	0.472	0.508
<i>Demographics</i>				
Foreign born	0.163	0.275	0.182	0.184
<i>Occupational skill group</i>				
Very high skilled	0.019	0.021	0.046	0.040
High skilled	0.044	0.047	0.079	0.070
Medium-high skilled	0.054	0.056	0.086	0.074
Medium-low skilled	0.590	0.608	0.636	0.647
Low skilled	0.292	0.267	0.153	0.168

Notes: The table presents key characteristics of workers in the construction sector for selected years. Sample restricted to workers aged 18–60 employed in construction in the reference year.

Table OA.2: First-Stage Diagnostics

	Joint F $F(2, 49)$	SW F $F(1, 49)$
Shock	6.24 [0.004]	12.51
Shock $\times$ Reallocation index	19.63 [0.000]	39.96
Anderson-Rubin $F(2, 49)$:	11.84	[0.000]

Notes: First-stage results for the IV specification with two endogenous variables (Shock and Shock $\times$ Reallocation index) instrumented by pre-recession construction growth (2000–2006) and its interaction with the reallocation index. Joint F tests excluded instruments jointly for each endogenous variable. SW F is the Sanderson–Windmeijer conditional statistics, which test instrument relevance for each endogenous variable separately. All statistics are cluster-robust (50 provinces). p -values in brackets.

Table OA.3: Sectoral mobility and the reallocation index, 2012

	(1)	(2)	(3)
	Sector Change, 2012		
Shock	0.206** (0.081)	0.211*** (0.075)	
Reall. Index		−0.045* (0.024)	
Shock $\times$ Reall. Index		0.120*** (0.037)	
<i>Shock $\times$ Reallocation Index Quartile:</i>			
Q1 (lowest)			0.073 (0.110)
Q2			0.075 (0.130)
Q3			0.243*** (0.090)
Q4 (highest)			0.452*** (0.078)
Observations	32,521	32,521	32,521
Controls	Yes	Yes	Yes

Notes: Linear probability estimates. Dependent variable: indicator for employment in a sector other than construction 2012. All specifications include the preferred set of individual and regional controls from Column (3) of Table 1. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.4: First Post-Displacement Sector by Reallocation Index Quartile

Destination sector	Reallocation index quartile				Total
	Q1	Q2	Q3	Q4	
Agriculture	40.7	31.7	17.7	9.8	100.0
Extractive ind.	20.4	19.7	25.2	34.7	100.0
Manufacturing	22.7	22.4	28.1	26.8	100.0
Energy & utilities	21.9	20.3	27.4	30.4	100.0
Commerce	28.0	24.9	27.2	20.0	100.0
Hospitality	33.6	28.6	24.6	13.2	100.0
Transport & comm.	21.3	23.2	30.7	24.8	100.0
Finance & insurance	13.1	19.3	25.3	42.4	100.0
Real estate	26.1	24.3	30.4	19.1	100.0
Professional svc.	19.9	19.1	26.4	34.6	100.0
Public admin./educ.	22.4	24.5	25.9	27.2	100.0
Other services	20.0	10.9	38.2	30.9	100.0
Total	25.4	24.7	25.4	24.5	100.0

Notes: Each row reports the distribution across reallocation index quartiles of workers whose first post-displacement job was in the indicated sector. Entries are row percentages: each row sums to 100. The reallocation index quartiles are defined over the full estimation sample.

Table OA.5: Conditional-Wage Null: Sensitivity to Employment Spell Threshold

	(1)	(2)	(3)
	p_{25} threshold	p_{50} threshold	p_{75} threshold
Shock	-0.005 (0.004)	-0.000 (0.004)	0.004 (0.005)
Shock $\times$ Reall. Index	0.003 (0.003)	-0.001 (0.003)	0.001 (0.004)
Observations	25,527	17,482	8,578
Controls	Yes	Yes	Yes

Notes: The dependent variable is average monthly earnings conditional on employment, normalized by the worker's average annual earnings in 2005–2006. Each column restricts the sample to workers whose cumulative employment days exceed the p_{25} , p_{50} , and p_{75} percentiles of the cumulative-days distribution, respectively. All specifications include worker, job, and regional controls from Column (3) of Table 1. Standard errors are clustered at the province level.

Table OA.6: Reallocation Index and Labor Market Outcomes, Controlling for Index Decile Fixed Effects

	(1)	(2)
	Cumulative employment	Cumulative earnings
Shock	−1.627*** (0.147)	−2.059*** (0.277)
Shock × Index	0.232*** (0.082)	0.211 (0.160)
Baseline controls	Yes	Yes
Index decile FE	Yes	Yes
Observations	34,000	34,000

Notes: Dependent variables are cumulative days employed annualized (Column 1) and normalized cumulative earnings (Column 2), each measured over 2007–2012 and normalized by the worker’s average annual earnings in 2005–2006. All specifications augment the baseline interaction from Table 3 with fixed effects for deciles of the reallocation index, flexibly absorbing any level relationship between the index and outcomes. Because the decile fixed effects condition on the index level nonparametrically, they remove more variation than the province fixed effects reported in the main text, which absorb aggregate shock severity and local wage levels rather than the index itself. Standard errors are clustered at the province level.

Table OA.7: Heterogeneity in Employment Losses: Shock Interactions

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Baseline effect</i>						
Shock	-1.161*** (0.210)	-1.580*** (0.154)	-1.170*** (0.151)	-1.319*** (0.209)	-1.847*** (0.222)	-1.127*** (0.212)
<i>Panel B: Interaction with worker characteristics</i>						
Shock $\times$ Low education	-0.612*** (0.207)					
Shock $\times$ Low tenure		-0.120 (0.271)				
Shock $\times$ Low experience			-0.861*** (0.164)			
Shock $\times$ Low earnings				-0.603** (0.242)		
Shock $\times$ Low skill					0.287 (0.252)	
Shock $\times$ Young						-0.972*** (0.206)
Shock $\times$ Middle age						-0.203 (0.225)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: Each column reports OLS estimates augmenting equation (2) with an interaction between the shock and an indicator for the low group within each dimension. Panel A reports the shock coefficient for the reference (high) group. Panel B reports the differential effect for the low group; the total effect for the low group is the sum of the two panels. Reference groups are: secondary education and above (Column 1), tenure above one year (Column 2), experience above six years (Column 3), above-median pre-recession earnings (Column 4), high-skilled (Column 5), and above 40 years of age (Column 6). Controls follow the same exclusion logic as Figure OA.14: the dimension being tested is excluded from the control set in the corresponding column. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.8: Heterogeneity in Cumulative Earnings Losses: Shock Interactions

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Baseline effect</i>						
Shock	-1.577*** (0.369)	-1.798*** (0.264)	-1.261*** (0.287)	-1.264*** (0.346)	-2.449*** (0.384)	-0.961*** (0.329)
<i>Panel B: Interaction with worker characteristics</i>						
Shock $\times$ Low education	-0.617** (0.295)					
Shock $\times$ Low tenure		-0.703* (0.381)				
Shock $\times$ Low experience			-1.838*** (0.282)			
Shock $\times$ Low earnings				-1.668*** (0.450)		
Shock $\times$ Low skill					0.491 (0.352)	
Shock $\times$ Young						-2.159*** (0.341)
Shock $\times$ Middle age						-0.514* (0.260)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: Each column reports OLS estimates augmenting equation (2) with an interaction between the shock and an indicator for the low group within each dimension. Panel A reports the shock coefficient for the reference (high) group. Panel B reports the differential effect for the low group; the total effect for the low group is the sum of the two panels. Reference groups are: secondary education and above (Column 1), tenure above one year (Column 2), experience above six years (Column 3), above-median pre-recession earnings (Column 4), high-skilled (Column 5), and above 40 years of age (Column 6). Controls follow the same exclusion logic as Figure OA.14: the dimension being tested is excluded from the control set in the corresponding column. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.9: Reallocation Index versus Outside-Options Index: A Head-to-Head

	Employment				Cumulative earnings	
	(1)	(2)	(3)	(4)	(5)	(6)
Shock	-1.634*** (0.160)	-1.609*** (0.155)	-1.615*** (0.158)	-1.634*** (0.163)	-2.075*** (0.284)	-2.072*** (0.294)
Shock $\times$ RI ^w	0.312*** (0.077)			0.383*** (0.085)	0.309* (0.159)	0.428** (0.178)
Shock $\times$ OOI ^{CD}		0.046 (0.091)		-0.169 (0.103)		-0.292* (0.170)
Shock $\times$ OOI ^{CD} (incl. const.)			0.030 (0.093)			
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: Both indices are standardized (mean zero, standard deviation one), so interaction coefficients are comparable in magnitude. RI^w is the wage-weighted reallocation index of Section 4. OOI^{CD} is a [Caldwell and Danieli \(2024\)](#) style outside-options index, constructed as the entropy of the worker's predicted sector distribution relative to the local (province-level) sectoral marginal, estimated from the 2006 cross-sectional stock and excluding wage information; the variant in column (3) includes construction in the opportunity set. The outcome in columns(1)–(4) is years-equivalent worked over 2007–2012; in columns (5)–(6), cumulative earnings over the same period. Columns (1)–(3) enter each interaction alone; column(4) includes RI^w and OOI^{CD} jointly. All specifications include the worker, job, and regional controls of Table 1. Standard errors clustered at the province level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.10: Province Block Bootstrap: Generated-Regressor Correction

	OLS (clustered)		Bootstrap (1,000 rep.)		Bootstrap CI	
	Coeff.	SE	Mean	SE	[p5, p95]	[p2.5, p97.5]
Employment	0.332	0.080	0.343	0.085	[0.216, 0.486]	[0.185, 0.521]
Cumul. earnings	0.344	0.162	0.331	0.153	[0.073, 0.579]	[0.004, 0.644]
Cond. wage	0.002	0.003	0.002	0.003	[-0.003, 0.006]	[-0.005, 0.007]

Notes: Province-level block bootstrap with 1,000 replications, resampling provinces jointly across the multinomial logit first stage and the second-stage OLS estimation. Each row reports the Shock $\times$ Reallocation Index interaction from the employment, cumulative earnings, and conditional wage specifications, respectively. OLS standard errors are clustered at the province level. The 90% and 95% bootstrap percentile intervals bound zero only for the conditional wage outcome.